

Integrating urban air mobility into the power grid through smart charging solutions

通过智能充电解决方案将城市空中交通融入电网

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ABSTRACT

摘要

Adapting the existing power grid to support large-scale urban air mobility (UAM) operations using electric vertical take-off and landing (eVTOL) aircraft presents a critical infrastructural challenge that needs to be tackled. To this end, this paper presents a framework for estimating the potential of smart charging to improve power system welfare when integrating large-scale UAM into the power grid. We first estimate passenger travel demand for UAM from location-based service (LBS) data. Then we obtain the feasible charging window of aircraft by solving a fleet dispatching problem to maximize the fulfillment of scheduled UAM travel demand. With consideration of power grid dynamics, we optimize the timing and rate of charging in each feasible charging window to maximize power system welfare without disturbing optimal fleet dispatch. Finally, uncertainties in arrival times and charging demands are managed with chance-constrained optimization. This ensures the failure probability for meeting charging demand stays below a set confidence level. The efficacy of the proposed approach was demonstrated by case studies using a 9-bus transmission system.

使现有的电网适应使用电动垂直起降 (eVTOL) 飞机来支持大规模城市空中交通 (UAM) 运营，是一项需要应对的关键基础设施挑战。为此，本文提出了一个框架，用于评估在将大规模 UAM 整合到电网中时，智能充电对提升电力系统效益的潜力。我们首先根据基于位置的服务 (LBS) 数据估算 UAM 的乘客出行需求。然后，通过解决机队调度问题以最大化满足预定的 UAM 出行需求，从而获得飞机的可行充电窗口。考虑到电网动态特性，我们在每个可行充电窗口中优化充电时间和充电速率，以在不干扰最佳机队调度的情况下最大化电力系统效益。最后，利用机会约束优化来管理到达时间和充电需求的不确定性。这确保了满足充电需求的失败概率保持在设定的置信水平以下。通过使用一个 9 节点输电系统的案例研究，证明了所提方法的有效性。

1. Introduction

1. 引言

The advancement of electric vertical take-off and landing (eVTOL) aircraft opens new opportunities for urban air mobility (UAM). UAM offers a novel solution to meet passengers' travel needs, facilitate goods delivery, and provide rapid response in case of emergencies. Moreover, it is considered a strategy for mitigating the escalating traffic congestion in major urban areas (Yan et al., 2024). Initially, the market held high expectations for UAM (Morgan Stanley Research, 2018). In addition to eVTOL companies like Joby Aviation (Joby Aviation, 2024), Uber initiated programs such as Uber Elevate to explore opportunities UAM (Uber, 2024). However, recent projections have been adjusted downward as various potential barriers have been identified (Morgan Stanley Research, 2021). These barriers include government intervention, public acceptance, and infrastructure limitations.

电动垂直起降 (eVTOL) 飞机的发展为城市空中交通 (UAM) 带来了新机遇。UAM 为满足乘客出行需求、促进货物配送以及在紧急情况下提供快速响应提供了一种新颖的解决方案。此外，它被视为缓解主要城市地区日益严重的交通拥堵的一项策略 (Yan 等人, 2024 年)。最初，市场对 UAM 寄予厚望 (摩根士丹利研究公司, 2018 年)。除了像 Joby Aviation 这样的 eVTOL 公司 (Joby Aviation, 2024 年)，优步还启动了诸如 Uber Elevate 等项目来探索 UAM 的机遇 (优步, 2024 年)。然而，由于发现了各种潜在障碍，最近的预测已向下调整 (摩根士丹利研究公司, 2021 年)。这些障碍包括政府干预、公众接受度和基础设施限制。

1.1. Background and motivation

1.1. 背景与动机

One of the infrastructure challenges that must be addressed for large-scale UAM operations comes from the power grid (Hirschberg, 2020). The success of UAM relies on the availability of electricity, as it involves the use of eVTOL aircraft. A reliable electricity supply is crucial for the daily operation of the eVTOL fleet, but the current power grid lacks the capacity for future

大规模城市空中交通 (UAM) 运营必须应对的基础设施挑战之一来自电网 (Hirschberg, 2020 年)。UAM 的成功依赖于电力供应，因为它涉及电动垂直起降 (eVTOL) 飞机的使用。可靠的电力供应对于 eVTOL 机队的日常运营至关重要，但当前电网缺乏应对未来 [此处原文不完整] 的能力

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Nomenclature

术语表

A

α, γ, ρ Function parameters estimated from data

α, γ, ρ 从数据中估计的函数参数

β_1, β_2 Dwell rate and burst rate

β_1, β_2 驻留率和突发率

δ_i^t Voltage angle in bus i at time slot t

δ_i^t 在时隙 t 时母线 i 中的电压角度

ℓ_i^t External load in bus i at time slot t

ℓ_i^t 在时隙 t 时母线 i 中的外部负载

ϵ Confidence parameter of the probabilistic constraint.

ϵ 概率约束的置信参数。

$\eta_{n,i}$ Binary value to represent whether session n is a load in bus i

$\eta_{n,i}$ 表示会话 n 是否为母线 i 中的负载的二进制值

$\mathcal{A}$ Set of aircrafts

$\mathcal{A}$ 飞机集合

I Set of buses in the power system

I 电力系统中的母线集合

$\mathcal{K}_n$ Set of all possible realizations of constraints of session n

$\mathcal{K}_n$ 会话 n 约束的所有可能实现的集合

$\mathcal{K}_n^+$ Set of all possible realizations of constraints of session n with a probability over the confidence parameter ϵ

$\mathcal{K}_n^+$ 会话 n 约束的所有可能实现的集合，其概率超过置信参数 ϵ

$\mathcal{K}_n^-$ Set of all possible realizations of constraints of session n with a probability less or equal the confidence parameter ϵ

$\mathcal{K}_n^-$ 会话 n 约束的所有可能实现的集合，其概率小于或等于置信参数 ϵ

$\mathcal{L}$ Set of factors that influence people's choice

$\mathcal{L}$ 影响人们选择的因素集合

$\mathcal{M}$ Set of vertiport charging stations

$\mathcal{M}$ 垂直机场充电站集合

$\mathcal{N}$ Set of charging sessions

$\mathcal{N}$ 充电会话集合

$\mathcal{R}$ Set of travel demand

$\mathcal{R}$ 出行需求集合

$\mathcal{T}$ Set of time slots

$\mathcal{T}$ 时隙集合

$\mathcal{V}$ Set of vertiports

$\mathcal{V}$ 垂直机场集合

$\mu_{v,v'}^t$ Energy consumption from vertiport v to vertiport v' at time slot t

$\mu_{v,v'}^t$ 在时隙 t 从垂直机场 v 到垂直机场 v' 的能耗

ω Flexibility time window

ω 灵活性时间窗口

Ω_n Joint Distribution of earliest start time, latest end time, and charging energy of session n

Ω_n 会话 n 最早开始时间、最晚结束时间和充电能量的联合分布

$\phi_{n,m}$ Binary value to represent whether session n is a load in station m

$\phi_{n,m}$ 表示会话 n 在站点 m 是否为负载的二进制值

$\pi_{n,k}$ Probability of sampled case k for session n

$\pi_{n,k}$ 会话 n 的采样案例 k 的概率

$\Psi_i(\cdot)$ Generation cost function of bus i

$\Psi_i(\cdot)$ 母线 i 的发电成本函数

$\tau_{n,k}^e$ Latest end time of session n in sampled case k

$\tau_{n,k}^e$ 采样案例 k 中会话 n 的最晚结束时间

$\tau_{n,k}^s$ Earliest start time of session n in sampled case k

$\tau_{n,k}^s$ 采样案例 k 中会话 n 的最早开始时间

τ_n^e Latest end time of session n

τ_n^e 会话 n 的最晚结束时间

τ_n^s Earliest start time of session n

τ_n^s 会话 n 的最早开始时间

$\Theta_i(\cdot)$ Generation benefit function of bus i

$\Theta_i(\cdot)$ 母线 i 的发电效益函数

$b_{i,j}$ Reactance of transmission line between bus i and bus j

$b_{i,j}$ 母线 i 和母线 j 之间输电线路的电抗

Cuam Passenger capacity per aircraft

每架飞机的城市空中交通乘客容量

$d_{n,k}$ Charging demand of session n in sampled case k

$d_{n,k}$ 采样案例 k 中会话 n 的充电需求

d_n Charging demand of session n

d_n 会话 n 的充电需求

$f_{i,j}^t$ Power flow of transmission line between bus i and bus j at time slot t

$f_{i,j}^t$ 在时隙 t 时, 母线 i 和母线 j 之间传输线的功率流

g_i^t Power generation in bus i at time slot t

g_i^t 在时隙 t 时, 母线 i 中的发电量

$h_{n,k}$ Binary variable to indicate if sampled case k for session n happens

$h_{n,k}$ 用于指示会话 n 的采样案例 k 是否发生的二元变量

N^{trips} Number of travel demand at specific vertiport and time interval

N^{trips} 在特定垂直机场和时间间隔的出行需求数量

n_w

$o_{v,v'}^t$ Traveling time from vertiport v to vertiport v' at time slot t

$o_{v,v'}^t$ 在时隙 t 时, 从垂直机场 v 到垂直机场 v' 的出行时间

$P^{exp}(\cdot)$ Probability function of an individual exploring a location with a specific rank

$P^{exp}(\cdot)$ 个体探索具有特定排名位置的概率函数

$P^{new}(\cdot)$ Probability function of an individual exploring a new location given the number of previously visited locations

$P^{new}(\cdot)$ 给定先前访问位置数量时, 个体探索新位置的概率函数

$P^{rhy}(\cdot)$ Fraction function of all trips of the population in a given time

$P^{rhy}(\cdot)$ 在给定时间内, 总体所有行程的分数函数

Probability function of an individual returning to the previously visited location

个体返回先前访问位置的概率函数

p_n^t Charging power of session n at time t

p_n^t 在时间 t 时, 会话 n 的充电功率

$q_{v,v'}^t$ Number of request from vertiport v to vertiport v' at time slot t

$q_{v,v'}^t$ 在时隙 t 时, 从垂直机场 v 到垂直机场 v' 的请求数量

$s_{n,k}^t$ Binary variable to indicate if the session n is allowed at time t in sampled case k

$s_{n,k}^t$ 用于指示在采样案例 k 中, 时间 t 是否允许会话 n 的二元变量

s_n^t Binary variable to indicate if the session n is allowed at time t

s_n^t 用于指示时间 t 是否允许会话 n 的二元变量

t^d, t^a Departure time and arrival time of a flight

t^d, t^a 航班的出发时间和到达时间

v^o, v^d Origin and destination vertiport

v^o, v^d 起始和目的地垂直机场

w_u^t Time to charge aircraft from SoC $u - 1$ to SoC u at time slot t

w_u^t 在时隙 t 将飞机从 SoC $u - 1$ 充电至 SoC u 所需的时间

$x_{v,u,u'}^t$ Number of aircraft at vertiport v that begin with SoC u and end with SoC u' at time slot t

$x_{v,u,u'}^t$ 在时隙 t ，在垂直起降机场 v 开始于 SoC u 并结束于 SoC u' 的飞机数量

$y_{v,v',u}^t$ Number of aircraft departing for vertiport v' from vertiport v with SoC u at time slot t

$y_{v,v',u}^t$ 在时隙 t ，从垂直起降机场 v 出发前往垂直起降机场 v' 且具有 SoC u 的飞机数量

$z_{v,u}^t$ Number of parked aircraft at vertiport v with SoC u at time slot t

$z_{v,u}^t$ 在时隙 t ，在垂直起降机场 v 具有 SoC u 的停放飞机数量

C_m^+ Maximum charging capacity at station m

C_m^+ 站点 m 的最大充电容量

d_n^+ Maximum overcharged demand of session n

d_n^+ 会话的最大过充电需求 n

d_n^- Maximum undercharged demand of session n

d_n^- 会话的最大欠充电需求 n

$F_{i,j}^+$ Maximum power flow of transmission line between bus i and bus j

$F_{i,j}^+$ 母线 i 和母线 j 之间传输线的最大功率流

$F_{i,j}^-$ Minimum power flow of transmission line between bus i and bus j

$F_{i,j}^-$ 母线 i 和母线 j 之间传输线的最小功率流

G_i^+ Maximum power generation in bus i

G_i^+ 母线 i 中的最大发电量

G_i^- Minimum power generation in bus i

G_i^- 母线 i 中的最小发电量

P^+ Maximum charging power

P^+ 最大充电功率

R_i^+ Upper bound of ramping limit of bus i

R_i^+ 母线 i 的爬坡限制上限

R_i^- Lower bound of ramping limit of bus i large-scale eVTOL charging (Connor O’Neil, 2024). Additionally, the increasing number of electric vehicles (EVs) worsens the strain on the power supply. The projection suggests EV percentage of the ground fleet in the U.S. will grow to about 50% in 2050 with higher charging rates greater than 300 kilowatts (kW) (BloombergNEF, 2023; Drive, 2019; Michelbacher et al., 2017; Plumer et al., 2021). The combination of more ground EVs and faster ground EV charging rates will utilize more of the electrical grid over the next several decades and thus reduce available electrical grid power capacity for UAM and constrain UAM operations if there are not enough upgrades of the power grid capacity.

R_i^- 母线 i 的爬坡限制下限大规模电动垂直起降飞行器充电 (康纳·奥尼尔 (Connor O’Neil), 2024 年)。此外, 电动汽车 (EV) 数量的增加加剧了电力供应的压力。预测表明, 到 2050 年, 美国地面车队中的电动汽车比例将增长至约 50%, 充电速率更高, 超过 300 千瓦 (kW)(彭博新能源财经 (BloombergNEF), 2023 年; Drive, 2019 年; 米歇尔巴赫等人 (Michelbacher et al.), 2017 年; 普卢默等人 (Plumer et al.), 2021 年)。在未来几十年中, 更多地面电动汽车与更快的地面电动汽车充电速率相结合, 将消耗更多的电网电力, 因此, 如果电网容量没有足够的升级, 将减少可用于城市空中交通 (UAM) 的电网电力容量, 并限制 UAM 的运营。

Under the best-case assumption that electricity can be transmitted and distributed as needed within each interconnection (i.e., electrical grid), Thippavong (2023) estimates the maximum number of UAM aircraft that can be charged simultaneously by dividing the available electrical grid power capacity for UAM by the baseline UAM aircraft charging rate. The results show that the maximum number of UAM aircraft that can charge simultaneously in the continental United States will decrease by 20% by 2050 due to the projected increase in ground EVs. However, achieving even this scenario could cost trillions in grid distribution (Marston, 2018) and transmission (Mackenzie, 2019) enhancements. Given the high cost of power network upgrades, an alternative solution is to intelligently coordinate aircraft charging. By aligning UAM charging with power system constraints, this coordination can minimize grid instability, such as overloads or voltage drops, without the need for costly infrastructure upgrades (Shareef et al., 2016; Richardson et al., 2011; Clement-Nyns et al., 2009). In this work, we aim to better understand the demand profiles of UAM fleets and develop smart charging strategies to optimally charge UAM fleets without overwhelming the electric grid.

在最佳情况假设下, 即电力可以在每个互联 (即电网) 内按需传输和分配, 蒂普哈冯 (Thippavong)(2023 年) 通过将可用于 UAM 的电网电力容量除以 UAM 飞机的基准充电速率, 估计了可以同时充电的 UAM 飞机的最大数量。结果表明, 由于预计地面电动汽车数量的增加, 到 2050 年, 美国大陆可以同时充电的 UAM 飞机的最大数量将减少 20%。然而, 即使实现这种情况, 也可能在电网配电 (马斯顿 (Marston), 2018 年) 和输电 (麦肯齐 (Mackenzie), 2019 年) 增强方面花费数万亿美元。鉴于电网升级成本高昂, 一种替代解决方案是智能协调飞机充电。通过使 UAM 充电与电力系统约束相匹配, 这种协调可以在无需昂贵的基础设施升级的情况下, 最大限度地减少电网不稳定, 如过载或电压降 (沙里夫等人 (Shareef et al.), 2016 年; 理查森等人 (Richardson et al.), 2011 年; 克莱门特 - 尼克斯等人 (Clement - Nyns et al.), 2009 年)。在这项工作中, 我们旨在更好地了解 UAM 机队的需求概况, 并制定智能充电策略, 以在不使电网不堪重负的情况下, 对 UAM 机队进行最优充电。

1.2. Related literature

1.2. 相关文献

Designing a smart charging schedule for the UAM fleet requires coordination between the power network and the transportation system. Controlling the UAM system involves two key tasks: (1) solving a dispatch problem to assign aircraft to fulfill trip requests and (2) monitoring the aircraft's state of charge (SoC) to manage the recharging of the aircraft battery. During this process, the electric UAM fleet represents a controllable load to the power grid in time and space. If well coordinated, this controllable load can improve grid social welfare even when considering power system operation constraints.

为城市空中交通 (UAM) 机队设计智能充电计划需要电力网络和运输系统之间的协调。控制 UAM 系统涉及两项关键任务:(1) 解决调度问题, 分配飞机以完成出行请求; (2) 监测飞机的充电状态 (SoC), 以管理飞机电池的充电。在此过程中, 电动 UAM 机队在时间和空间上对电网而言是一个可控负载。如果协调得当, 即使考虑电力系统运行约束, 这个可控负载也可以提高电网的社会福利。

There is a growing body of work in UAM fleet dispatch. For example, Shihab et al. developed a model to decide whether to offer on-demand or scheduled flights and how to dispatch the fleet and schedule operations based on simulated market demand (Shihab et al., 2019). Munari et al. assigned aircraft to on-demand requests while accounting for maintenance events, allowing flight upgrades to reduce operational and repositioning costs (Munari and Alvarez, 2019). Stouffer et al. designed a dispatching tool for UAM operations that enables a dispatcher to plan a UAM flight and check for issues before filing a flight plan (Stouffer and Kostiuk, 2020). Besides the fleet dispatch, another challenge comes from the charging strategies. Optimal charging strategies for fleet aircraft have received a lot of attention given the high cost of batteries for aircraft, several studies have examined how different recharging strategies can be optimized to help regulate the charge profile and enhance battery life (Amjad et al., 2018; Sweda et al., 2017; Wu and Sioshansi, 2017; Widrick et al., 2018).

在 UAM 机队调度方面的工作越来越多。例如, Shihab 等人开发了一个模型, 用于决定是提供按需航班还是定期航班, 以及如何根据模拟市场需求调度机队和安排运营 (Shihab 等人, 2019 年)。Munari 等人在考虑维护事件的情况下, 为按需请求分配飞机, 允许航班升级以降低运营和重新定位成本 (Munari 和 Alvarez, 2019 年)。Stouffer 等人设计了一个用于 UAM 运营的调度工具, 使调度员能够规划 UAM 航班并在提交飞行计划前检查问题 (Stouffer 和 Kostiuk, 2020 年)。除了机队调度, 另一个挑战来自充电策略。鉴于飞机电池成本高昂, 机队飞机的最优充电策略受到了很多关注, 一些研究探讨了如何优化不同的充电策略, 以帮助调节充电曲线并延长电池寿命 (Amjad 等人, 2018 年; Sweda 等人, 2017 年; Wu 和 Sioshansi, 2017 年; Widrick 等人, 2018 年)。

While previous studies have examined the UAM fleet as part of the transportation system, they often focus solely on minimizing charging costs and overlook the impact on the power grid. To address this gap, some research has included the power system in the analysis. For example, Shihab et al. developed an optimal fleet dispatch framework for eVTOL aircraft aiming to boost fleet operators' revenue by securing extra income through offering frequency regulation services to the electricity market (Shihab et al., 2020). Thippavong documented an analysis of the available electrical grid power capacity for UAM (Thippavong, 2022, 2023). Additionally, Acera et al. proposed an optimization based on genetic algorithms to determine the number of eVTOL vehicles required to make vehicle-to-grid services profitable for different electrical systems (Velaz-Acera et al., 2023). Paudyal et al. conducted a grid impact study and analyzed infrastructure upgrade require-

ments for eVTOL charging on a realistic distribution grid through power flow simulation analysis (Paudyal et al., 2024). However, these studies do not take into account, from the viewpoint of the power grid operator, how the smart charging of the eVTOL aircraft fleet could benefit the grid considering power flow dynamics. In this work, we aim to evaluate the benefits to the power grid by implementing smart charging schedules that consider grid constraints.

虽然之前的研究将 UAM 机队视为运输系统的一部分，但它们通常只专注于最小化充电成本，而忽略了对电网的影响。为了弥补这一差距，一些研究在分析中纳入了电力系统。例如，Shihab 等人开发了一个用于电动垂直起降 (eVTOL) 飞机的最优机队调度框架，旨在通过向电力市场提供频率调节服务来获取额外收入，从而提高机队运营商的收入 (Shihab 等人, 2020 年)。Thippavong 记录了对 UAM 可用电网电力容量的分析 (Thippavong, 2022 年, 2023 年)。此外，Acera 等人提出了一种基于遗传算法的优化方法，以确定使车辆到电网服务在不同电力系统中盈利所需的 eVTOL 车辆数量 (Velaz - Acera 等人, 2023 年)。Paudyal 等人进行了电网影响研究，并通过潮流模拟分析，分析了在实际配电网中 eVTOL 充电的基础设施升级要求 (Paudyal 等人, 2024 年)。然而，从电网运营商的角度来看，这些研究没有考虑 eVTOL 机队的智能充电如何在考虑潮流动态的情况下使电网受益。在这项工作中，我们旨在通过实施考虑电网约束的智能充电计划来评估对电网的益处。

Most research on dispatching and scheduling algorithms for UAM have used deterministic frameworks. However, these models overlook the stochastic nature of aircraft charging (Federal Aviation Administration, 2023). This stochasticity is influenced by weather, temperature, and emergencies. As a result, deterministic models lead to ineffective schedules. Recent studies have started to address uncertainties in UAM operations. They consider uncertainty in deconfliction (Wu et al., 2022; Evans et al., 2021), dispatch (Wei et al., 2021; Paul et al., 2024), fleet sizing (Cao et al., 2024), routing (Wei et al., 2023), traffic flow management (Pang et al., 2024), and aircraft control (Pradeep et al., 2021). However, these studies do not consider the impact of UAM operation uncertainty on the power grid. In this work, we use a chance-constrained optimization approach. This approach models the stochastic characteristics of aircraft operations using mixed-integer programming (MIP). The chance constraints ensure the failure probability of the UAM charging plan stays below a predetermined confidence level. Our objective is to improve the overall welfare of the power system by considering uncertainties in arrival timings and charging demands while not disturbing the optimal dispatch of eVTOL aircraft fleets that maximize the fulfillment of UAM demands.

大多数关于 UAM 调度和排班算法的研究都使用了确定性框架。然而，这些模型忽略了飞机充电的随机性 (美国联邦航空管理局, 2023 年)。这种随机性受天气、温度和突发事件的影响。因此，确定性模型导致的调度无效。最近的研究开始解决 UAM 运营中的不确定性。它们考虑了冲突解决 (Wu 等人, 2022 年; Evans 等人, 2021 年)、调度 (Wei 等人, 2021 年; Paul 等人, 2024 年)、机队规模确定 (Cao 等人, 2024 年)、航线规划 (Wei 等人, 2023 年)、交通流管理 (Pang 等人, 2024 年) 和飞机控制 (Pradeep 等人, 2021 年) 中的不确定性。然而，这些研究没有考虑 UAM 运营不确定性对电网的影响。在这项工作中，我们使用机会约束优化方法。这种方法使用混合整数规划 (MIP) 对飞机运营的随机特性进行建模。机会约束确保 UAM 充电计划的失败概率保持在预定的置信水平以下。我们的目标是通过考虑到到达时间和充电需求的不确定性来提高电力系统的整体福利，同时不干扰使 UAM 需求得到最大满足的 eVTOL 飞机机队的最优调度。

1.3. Our contribution

1.3. 我们的贡献

We propose a framework to estimate the potential of smart charging to improve power system welfare when integrating large-scale UAM into the power grid. The framework begins with the operator estimating UAM travel demand using location-based service (LBS) data. Based on this estimated demand, the system operator determines fleet dispatch and models the impact of fleet charging on the power system by solving a DC optimal power flow problem. To account for uncertainties in arrival times and charging demands, we use a chance-constrained formulation. The framework is shown in Fig. 1. The contribution of the work can be categorized into three main points:

我们提出了一个框架，用于估计在将大规模城市空中交通 (UAM) 整合到电网时，智能充电对改善电力系统福利的潜力。该框架首先由运营商使用基于位置的服务 (LBS) 数据估计 UAM 出行需求。基于此估计需求，系统运营商确定机队调度，并通过求解直流最优潮流问题来模拟机队充电对电力系统的影响。为了考虑到达时间和充电需求的不确定性，我们使用机会约束公式。该框架如图 1 所示。这项工作的贡献可以分为三个要点：

- **Travel Demand Estimation:** We first estimate the city-level mobility pattern using Location-Based Service (LBS) data and the TimeGeo model. Then we estimate the UAM trips given an UAM penetration level based on this city-level travel demand.

- **出行需求估计:** 我们首先使用基于位置的服务 (LBS) 数据和 TimeGeo 模型估计城市层面的出行模式。然后，我们根据此城市层面的出行需求，在给定 UAM 渗透率水平的情况下估计 UAM 出行。

- **Transportation and Power System Nexus:** We calculate an optimal fleet dispatch policy with integer programming (IP) to maximize the fulfillment of day-ahead scheduled UAM travel demand. Then, we model the fleet's impact on the power network with the DC optimal flow model. The goal is to maximize power system welfare through smart charging scheduling without changing fleet dispatch.

- **交通与电力系统关联:** 我们使用整数规划 (IP) 计算最优机队调度策略，以最大限度地满足提前一天安排的 UAM 出行需求。然后，我们用直流最优潮流模型模拟机队对电网的影响。目标是通过智能充电调度最大化电力系统福利，而不改变机队调度。

- **Stochastic Characterization:** We consider the uncertainty in UAM operations and propose charging schedules using chance-constrained MIP. We compare scenarios with and without uncertainty. This comparison helps us evaluate how uncertainty affects the effectiveness of smart charging in mitigating the negative impact of large-scale UAM on the power grid.

- **随机特征描述:** 我们考虑 UAM 运营中的不确定性，并使用机会约束混合整数规划提出充电调度。我们比较有无不确定性的情景。这种比较有助于我们评估不确定性如何影响智能充电在减轻大规模 UAM 对电网负面影响方面的有效性。

The rest of the paper is organized as follows. Section 2 demonstrates the method for UAM travel demand estimation. Section 3 introduces the fleet dispatch algorithm. Section 4 provides both deterministic

and stochastic frameworks to estimate the impact of the UAM fleet on the power grid under smart scheduling. Section 6 presents numerical studies and Section 7 concludes the paper.

本文的其余部分组织如下。第 2 节展示了 UAM 出行需求估计的方法。第 3 节介绍了机队调度算法。第 4 节提供了确定性和随机框架，以估计智能调度下 UAM 机队对电网的影响。第 6 节给出了数值研究，第 7 节总结了本文。

2. UAM travel demand estimation

2. 城市空中交通出行需求估计

In this section, we first introduce the mobility generation model, then we illustrate how we estimate the UAM travel demand from the generated city-wide mobility trajectories.

在本节中，我们首先介绍出行生成模型，然后说明如何从生成的全市出行轨迹中估计城市空中交通的出行需求。

2.1. Mobility generation framework: TimeGeo

2.1. 出行生成框架: 时间地理模型

We use TimeGeo (Jiang et al., 2016) to generate individual trajectories for the entire city population. The purpose is to generate the urban-scale Origin-Destination (OD) travel flow from which we can estimate the demand for UAM services.

我们使用时间地理模型 (Jiang 等人, 2016 年) 为整个城市人口生成个体轨迹。目的是生成城市规模的起讫点 (OD) 出行流量，以便我们能够估计对城市空中交通 (UAM) 服务的需求。

TimeGeo models each individual's trajectory based on temporal and spatial choices: Temporal choices determine how long to stay and when to leave, while spatial choices decide which places to visit. Since the data covers only a portion of the population, we need to scale it to represent the entire population. The details on modeling these choices and expanding the data are provided below.

时间地理模型基于时间和空间选择对每个人的轨迹进行建模: 时间选择决定停留时长和离开时间，而空间选择决定前往哪些地点。由于数据仅涵盖部分人口，我们需要对其进行缩放以代表整个人口。以下提供了对这些选择进行建模和扩展数据的详细信息。

To model temporal choice, TimeGeo's time-inhomogeneous Markov chain model defines three stay states: home, work, or other. It models work activities with predetermined arrival times and durations. Home is considered a less active state than other due to longer average stay durations.

为了对时间选择进行建模, 时间地理模型的时间非齐次马尔可夫链模型定义了三种停留状态: 家、工作地点或其他地点。它使用预定的到达时间和持续时间对工作活动进行建模。由于平均停留时间较长, 家被认为是比其他地点活跃度更低的状态。

For individuals at home, the probability of starting a trip within a specific time interval t is denoted as $n_w P^{rhy}(t)$. Here, n_w , quantifies the number of trips originating from home, and $P^{rhy}(t)$ reflects the population's overall travel circadian rhythm. The framework distinguishes between $P^{rhy}(t)$ for commuters and non-commuters. For non-commuters, $P^{rhy}(t)$ is the fraction of all trips of the population in the interval t . For commuters, since work is modeled as a fixed activity, $P^{rhy}(t)$ does not include trips to or from work. The product $n_w P^{rhy}(t)$ defines the individual's travel probability at a specific time interval t when they are at home.

对于在家中的个体, 在特定时间间隔 t 内开始出行的概率表示为 $n_w P^{rhy}(t)$ 。这里, n_w 量化了从家出发的出行次数, $P^{rhy}(t)$ 反映了人口的整体出行昼夜节律。该框架区分通勤者和非通勤者的 $P^{rhy}(t)$ 。对于非通勤者, $P^{rhy}(t)$ 是该时间段内人口所有出行的比例。对于通勤者, 由于工作被建模为固定活动, $P^{rhy}(t)$ 不包括上下班的出行。乘积 $n_w P^{rhy}(t)$ 定义了个体在家时在特定时间间隔 t 的出行概率。

For individuals at other, the intention to initiate a trip is modeled with two parameters: dwell rate β_1 and burst rate β_2 . Dwell rate β_1 measures how much more active the person is likely to travel when in state other compared to state home. Burst rate β_2 measures how much more active the person is likely to travel to an additional other location when in state other compared to state home. Specifically, the probability of traveling, when an individual is at state other, is defined as $\beta_1 n_w P^{rhy}(t)$. Next, if this individual chooses to move at time t , TimeGeo defines the probability that an individual travels to an additional other location as $\beta_2 n_w P^{rhy}(t) \beta_1 n_w P^{rhy}(t)$.

对于在其他地点的个体, 出行意图用两个参数建模: 停留率 β_1 和突发率 β_2 。停留率 β_1 衡量个体在处于“其他地点”状态时比处于“家”状态时出行可能性高多少。突发率 β_2 衡量个体在处于“其他地点”状态时前往另一个额外的“其他地点”的可能性比处于“家”状态时高多少。具体而言, 个体处于“其他地点”状态时出行的概率定义为 $\beta_1 n_w P^{rhy}(t)$ 。接下来, 如果该个体在时间 t 选择移动, 时间地理模型将个体前往另一个额外的“其他地点”的概率定义为 $\beta_2 n_w P^{rhy}(t) \beta_1 n_w P^{rhy}(t)$ 。

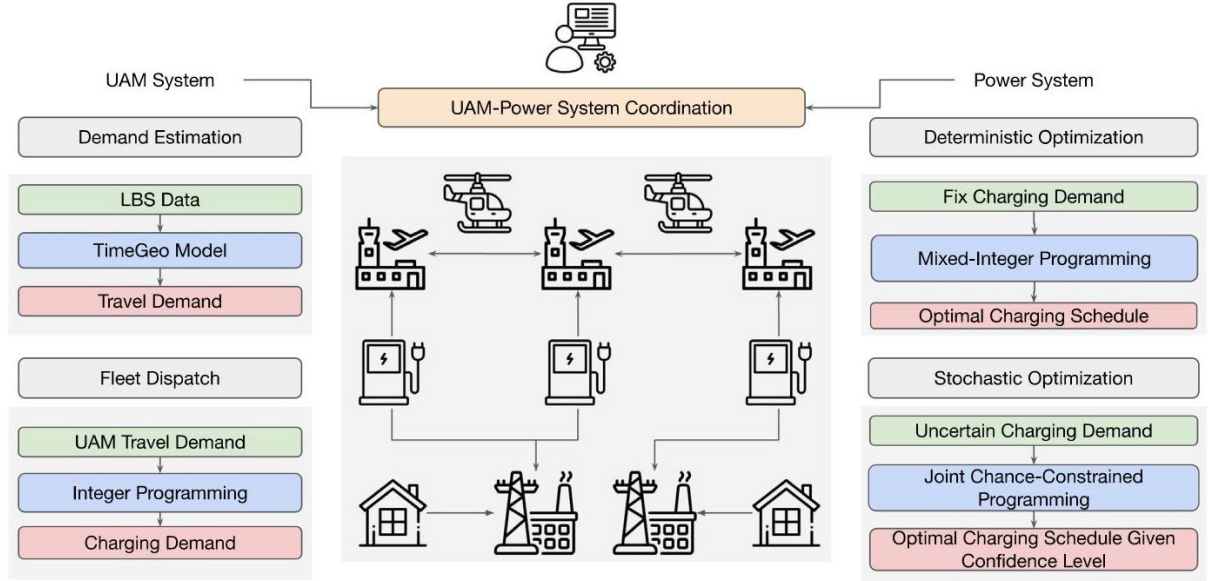


Fig. 1. Framework.

图 1. 框架

To model spatial choice, TimeGeo uses the rank-based exploration and preferential return (r-EPR) model. The temporal choice determines when to visit home, work, and other. While home and work are unique places, there are many possible other candidates. In spatial choice modeling, the framework aims to determine which other place to visit for each individual.

为了对空间选择进行建模，时间地理模型使用基于排名的探索和优先返回 (r-EPR) 模型。时间选择决定何时前往家、工作地点和其他地点。虽然家和工作地点是特定地点，但有许多可能的“其他地点”候选。在空间选择建模中，该框架旨在为每个个体确定前往哪个“其他地点”。

If the individual's temporal choice is to go to an other state, she explores a new location with probability P^{new} or returns to a previously visited location with probability $1 - P^{\text{new}}$. Denote λ as the number of previously visited locations, the P^{new} is defined as:

如果个体的时间选择是前往“其他地点”状态，她以概率 P^{new} 探索新地点或以概率 $1 - P^{\text{new}}$ 返回之前去过的地点。将 λ 表示为之前去过的地点数量， P^{new} 定义为：

$$P^{\text{new}}(\lambda) = \rho\lambda^{-\gamma}, \quad (1)$$

where ρ and γ are fitted from empirical data. If the individual decides to return to previously visited locations, she chooses a specific location l with probability $P^{\text{rtn}}(l)$. This probability is defined as the visitation frequency of location l . If the individual decides to explore a new location, she selects a destination based on the rank of locations. This rank represents the distance between each alternative destination and the current location. The shorter the distance, the higher the rank. Specifically, they choose a location l with rank κ with probability $P^{\text{exp}}(\kappa)$:

其中 ρ 和 γ 根据经验数据拟合。如果个体决定返回之前去过的地点，她以概率 $P^{rtn}(l)$ 选择特定地点 l 。此概率定义为地点 l 的访问频率。如果个体决定探索新地点，她根据地点的排名选择目的地。此排名表示每个替代目的地与当前位置之间的距离。距离越短，排名越高。具体而言，他们以概率 $P^{exp}(\kappa)$ 选择排名为 κ 的地点 l ：

$$P^{exp}(\kappa) \propto \kappa^{-\alpha}, \quad (2)$$

where α is the fitted parameter from empirical data.

其中 α 是根据经验数据拟合的参数。

By combining temporal and spatial choices with estimated parameters, we simulate mobility for users in the dataset with spatial resolution of 400 m and temporal resolution of 10 min . Since the dataset represents only a sample of the entire population, the next step is to expand these users to represent the whole population for a city-wide simulation. TimeGeo calculates two sets of expansion factors to scale commuters and non-commuters to match the real population. These factors determine how many times to replicate each commuter and non-commuter within each geographical unit (e.g., census tract, county).

通过将时间和空间选择与估计参数相结合，我们以 400 m 的空间分辨率和 10 min 的时间分辨率模拟数据集中用户的出行情况。由于该数据集仅代表整个人口的一个样本，下一步是扩展这些用户以代表整个城市人口进行全市范围的模拟。TimeGeo 计算两组扩展因子，以对通勤者和非通勤者进行缩放，使其与实际人口相匹配。这些因子决定了在每个地理单元（例如人口普查区、县）内每个通勤者和非通勤者要复制的次数。

2.2.UAM demand

2.2. 城市空中交通需求

The simulated demand gives a holistic representation of the travel pattern in the city. However, not all of the demand will be satisfied by UAM. This section demonstrates a method to predict which trips will likely be served by UAM at a given penetration level.

模拟需求全面反映了城市中的出行模式。然而，并非所有需求都能由城市空中交通满足。本节展示了一种方法，用于预测在给定渗透率水平下，哪些出行可能由城市空中交通提供服务。

Let the set of factors influencing people's choices be denoted as $\mathcal{L}$. Those factors are ordered based on importance, producing a factor vector $\mathbf{L} = (L_1, L_2, \dots, L_m)$, where $L_1 \in \mathcal{L}$ is the most important factor, and $L_m \in \mathcal{L}$ is the least important. Examples of factors that significantly influence people's decisions to choose eVTOL can be broadly classified into three categories, namely trip-related factors (e.g., travel time, cost, and distance), surface traffic congestion as a motivating factor to UAM, and acceptance-related factors (e.g., safety and security, privacy, and noise) (Long et al., 2023). In this work, we only consider trip-related factors due to data availability. However, the motivating factor and acceptance-related factors can be easily incorporated into the model once data is available. For each trip t , we compute its rank $r_L(t)$ with respect to each factor L . For example, if we define L as saved travel time, for a trip t_1 with a 30 min saved travel time and a trip t_2 with

a 20 min saved travel time, we have $r_L(t_1) = 1$ and $r_L(t_2) = 2$. The synthetic rank of trip t is then represented as a tuple:

设影响人们选择的因素集表示为 $\mathcal{L}$ 。这些因素根据重要性排序，生成一个因素向量 $\mathbf{L} = (L_1, L_2, \dots, L_m)$ ，其中 $L_1 \in \mathcal{L}$ 是最重要的因素， $L_m \in \mathcal{L}$ 是最不重要的因素。显著影响人们选择电动垂直起降飞行器决策的因素示例大致可分为三类，即出行相关因素(如出行时间、成本和距离)、作为城市空中交通激励因素的地面交通拥堵，以及接受相关因素(如安全保障、隐私和噪音)(Long 等人，2023 年)。在本研究中，由于数据可用性，我们仅考虑出行相关因素。然而，一旦有数据可用，激励因素和接受相关因素可以很容易地纳入模型。对于每次出行 t ，我们计算其相对于每个因素 L 的排名 $r_L(t)$ 。例如，如果我们将 L 定义为节省的出行时间，对于一次节省 30 min 出行时间的出行 t_1 和一次节省 20 min 出行时间的出行 t_2 ，我们有 $r_L(t_1) = 1$ 和 $r_L(t_2) = 2$ 。出行 t 的综合排名然后表示为一个元组：

$$\mathbf{r}(t) = (r_{L_1}(t), r_{L_2}(t), \dots, r_{L_m}(t)), \quad (3)$$

where $r_{L_i}(t)$ denotes the rank of trip t based on factor L_i , with L_1 being the most important factor and L_m the least important. The tuple is sorted lexicographically to prioritize trips based on their ranks for each factor, with the most important factors influencing the order first. We then select the top N^{trips} trips after lexicographic sorting of all $\mathbf{r}(t)$. The core of this method is to rank all trips by their likelihood of being UAM trips and selecting the top ones as UAM trips given a penetration level (refer to Appendix for an example).

其中 $r_{L_i}(t)$ 表示出行 t 基于因素 L_i 的排名， L_1 是最重要的因素， L_m 是最不重要的因素。对该元组按字典顺序排序，以便根据每个因素的排名对出行进行优先级排序，最重要的因素首先影响排序。然后，在对所有 $\mathbf{r}(t)$ 进行字典排序后，我们选择排名靠前的 N^{trips} 次出行。该方法的核心是根据所有出行成为城市空中交通出行的可能性对其进行排名，并在给定渗透率水平下选择排名靠前的出行作为城市空中交通出行(示例请参考附录)。

3. Aircraft fleet dispatch

3. 机队调度

In this section, we followed (Cao et al., 2024) to calculate the optimal day-ahead aircraft dispatch and charging window to maximize the satisfied requests while considering the system dynamics.

在本节中，我们遵循 (Cao 等人，2024 年) 的方法，在考虑系统动态的情况下，计算最优的提前一天飞机调度和充电窗口，以最大化满足的请求。

3.1. System model

3.1. 系统模型

This system has three components: vertiports, aircraft, and travel demand requests. The operator manages several vertiports and aircraft. Each aircraft can land at or take off from a vertiport. Upon landing, an

aircraft can choose to charge its battery at the vertiport. Each vertiport features a charging station equipped with several charging piles capable of accommodating multiple simultaneous charging sessions. We assume the fleet size and the number of vertiports are fixed, so not all travel demand can be satisfied. Passengers needing a UAM ride send a request a day in advance (i.e., the operator knows the demand for each time slot and each vertiport day-ahead), and the operator decides which requests to accept and how to dispatch aircraft among vertiports. For convenience, let $\mathcal{T}$ represent the discretized decision time intervals of the daily operation.

该系统有三个组件: 垂直起降机场、飞机和出行需求请求。运营商管理多个垂直起降机场和飞机。每架飞机可以在垂直起降机场降落或起飞。降落时, 飞机可以选择在垂直起降机场为其电池充电。每个垂直起降机场都设有一个充电站, 配备有多个充电桩, 能够同时容纳多个充电会话。我们假设机队规模和垂直起降机场的数量是固定的, 因此并非所有出行需求都能得到满足。需要城市空中交通出行的乘客提前一天发送请求(即, 运营商提前知道每个时间段和每个垂直起降机场的需求), 然后运营商决定接受哪些请求以及如何垂直起降机场之间调度飞机。为方便起见, 设 $\mathcal{T}$ 表示每日运营的离散决策时间间隔。

Vertiport. Denote $\mathcal{V}$ as the set of vertiports in a city where aircraft travel. Let $\mathcal{M}$ be the set of charging stations equipped at vertiports. Ideally, we need to constrain the number of chargers and parking spaces. However, in this work, we relax this constraint. This is because currently we are at a very early stage of UAM deployment and there is not so much infrastructure to give us a proper assumption of how many spaces are needed at each vertiport. In this regard, this formulation treats the infrastructure requirements as optimization variables. The results reveal what the ideal space each vertiport needs if we want to maximize the number of served trips. Let $o_{v,v'}$ denote the time cost between vertiports v and v' . Similarly, let $\mu_{v,v'}^t$ represent the energy cost between vertiports v and v' at time slot t . The time and energy required do not change with the number of passengers on the aircraft. However, these costs are different for each flight, as they are determined by the distance between departure and arrival vertiports along with the energy consumption per mile. The per-mile energy consumption varies due to external conditions that change throughout the day.

垂直起降机场。将 $\mathcal{V}$ 表示为飞机飞行所在城市的垂直起降机场集合。设 $\mathcal{M}$ 为垂直起降机场配备的充电站集合。理想情况下, 我们需要限制充电器数量和停车位数量。然而, 在这项工作中, 我们放宽了这一限制。这是因为目前我们正处于城市空中交通部署的非常早期阶段, 没有太多基础设施可供我们对每个垂直起降机场需要多少空间做出恰当假设。在这方面, 此公式将基础设施要求视为优化变量。结果揭示了如果我们想最大化服务出行次数, 每个垂直起降机场理想的的需求空间是多少。设 $o_{v,v'}$ 表示垂直起降机场 v 和 v' 之间的时间成本。类似地, 设 $\mu_{v,v'}^t$ 表示在时间段 t 垂直起降机场 v 和 v' 之间的能量成本。所需的时间和能量不会随飞机上的乘客数量而变化。然而, 每次飞行的这些成本是不同的, 因为它们由出发和到达垂直起降机场之间的距离以及每英里的能量消耗决定。每英里的能量消耗会因全天变化的外部条件而有所不同。

Aircraft. Denote $\mathcal{A}$ as the set of aircraft. Initially, all aircraft are assigned an initial vertiport according to fleet dispatch optimization results. During the time window after the arrival time of the current request and before the departure time of the next request, the aircraft can be charged to ensure operation. Note that aircraft is allowed not to charge if it does not increase the number of satisfied trips. At each time slot, one aircraft can be in one of three states: parked at vertiport, charging at vertiport, or in flight. We discrete the target SoC into U levels. To charge the aircraft from SoC $u - 1$ to SoC u at time t , the time cost is w_u^t . Each charging session has a predefined time window ω to provide flexibility in operation uncertainty.

飞机。将 $\mathcal{A}$ 表示为飞机集合。最初，根据机队调度优化结果为所有飞机分配一个初始垂直起降机场。在当前请求到达时间之后且下一个请求出发时间之前的时间窗口内，飞机可以充电以确保运行。请注意，如果不增加满足的出行次数，飞机可以不充电。在每个时间段，一架飞机可以处于三种状态之一：停在垂直起降机场、在垂直起降机场充电或飞行中。我们将目标荷电状态离散为 U 个级别。要在时间 t 将飞机从荷电状态 $u - 1$ 充电到荷电状态 u ，时间成本为 w_u^t 。每个充电会话都有一个预定义的时间窗口 ω ，以在运行不确定性方面提供灵活性。

Request. Denote $\mathcal{R}$ as the set of travel demand requests. For computational ease, we assume all users have "car-pooling" as the primary mode. That is, all users are comfortable taking rides with others. The pool rides are only limited to customers who share the same origin and destination. In this context, we combine individual requests N^{trips} into group requests. Every group request accommodates C^{uam} trips. An aircraft is capable of handling only one group request at a time. The number of requests from vertiport v to vertiport v' at time t is $q_{v,v'}^t$. Each request $r \in \mathcal{R}$ is represented by a tuple $\langle t^d, t^a, v^o, v^d \rangle$, where $t^d \in \mathcal{T}$ is the departure time, $t^a \in \mathcal{T}$ is the arrival time, $v^o \in \mathcal{V}$ is the origin vertiport, and v^d is the destination vertiport.

请求。将 $\mathcal{R}$ 表示为出行需求请求集合。为便于计算，我们假设所有用户都以“拼车”为主要模式。也就是说，所有用户都愿意与他人同乘。拼车仅限于共享相同出发地和目的地的客户。在这种情况下，我们将单个请求 N^{trips} 合并为组请求。每个组请求容纳 C^{uam} 次出行。一架飞机一次只能处理一个组请求。在时间 t 从垂直起降机场 v 到垂直起降机场 v' 的请求数量为 $q_{v,v'}^t$ 。每个请求 $r \in \mathcal{R}$ 由一个元组 $\langle t^d, t^a, v^o, v^d \rangle$ 表示，其中 $t^d \in \mathcal{T}$ 是出发时间， $t^a \in \mathcal{T}$ 是到达时间， $v^o \in \mathcal{V}$ 是出发垂直起降机场， v^d 是目的地垂直起降机场。

3.2. Decision variables

3.2. 决策变量

For each time slot t , denote the number of aircraft at vertiport v that begin to charge with an initial SoC of u and target SoC of u' as an integer variable $x_{v,u,u'}^t$. The number of aircraft departing for vertiport v' from vertiport v with SoC u is $y_{v,v',u}^t$. The number of parked aircraft at vertiport v of SoC u is $z_{v,u}^t$. We add one auxiliary integer variable $t_{v,v'}^t$ to represent the number of unserved travel demand.

对于每个时隙 t ，将在垂直起降机场 v 开始充电且初始荷电状态为 u 、目标荷电状态为 u' 的飞机数量表示为整数变量 $x_{v,u,u'}^t$ 。从垂直起降机场 v 出发前往垂直起降机场 v' 且荷电状态为 u 的飞机数量为 $y_{v,v',u}^t$ 。在垂直起降机场 v 且荷电状态为 u 的停放飞机数量为 $z_{v,u}^t$ 。我们添加一个辅助整数变量 $t_{v,v'}^t$ 来表示未满足的出行需求数量。

3.3. Objective

3.3. 目标

We aim to minimize the number of requests denied by the UAM operator:

我们的目标是尽量减少城市空中交通 (UAM) 运营商拒绝的请求数量:

$$\sum_{v,v',t} t_{v,v'}^t. \quad (4)$$

3.4. Constraints

3.4. 约束条件

Lower Bound of SoC to Start Flight. Ensure that aircraft cannot be in flight when the level reaches reserve SoC.

起飞时的电池荷电状态下限。确保当电量水平达到储备电池荷电状态时飞机不能处于飞行状态。

$$y_{v,v',0}^t = 0, \forall v, v', t. \quad (5)$$

Non-negative Unserved Orders. Ensure that the auxiliary integer variable $t_{p,p'}^t$ represents the number of unserved trips between vertiports. This does not prevent the dead heading as $q_{v,v'}^t - \sum_u y_{v,v',u}^t$ can be negative when there is repositioning flights.

非负未服务订单。确保辅助整数变量 $t_{p,p'}^t$ 表示垂直起降机场之间未服务行程的数量。这并不妨碍空载飞行，因为在有重新定位航班时 $q_{v,v'}^t - \sum_u y_{v,v',u}^t$ 可以为负。

$$t_{v,v'}^t \geq 0, \forall v, v', t \quad (6)$$

$$t_{v,v'}^t \geq q_{v,v'}^t - \sum_u y_{v,v',u}^t, \forall v, v', t. \quad (7)$$

Fleet size Limit. Ensures that the total number of aircraft in the system at the beginning equals the fleet size.

机队规模限制。确保系统开始时飞机的总数等于机队规模。

$$\sum_{v,u,u'} x_{v,u,u'}^0 + \sum_{v,u,u'} x_{v,u,u'}^0 + \sum_{v,u} z_{v,u}^0 = |\mathcal{A}|, \forall v, v'. \quad (8)$$

System Dynamics. Ensure the number of aircraft in parked state at each vertiport at each time slot at each SoC level is determined by parked aircraft from the previous time slot, arrivals and departures in the vertiport, and charging status of aircraft in the vertiport. Denote $\mathcal{W}_{v,v'}^t = \{t' \in \mathcal{T} \mid t' + o_{v,v'}^{t'} = t\}$, the constraint is:

系统动态特性。确保在每个时间槽、每个电池荷电状态水平下，每个垂直起降机场处于停放状态的飞机数量由上一个时间槽的停放飞机、垂直起降机场的起降情况以及垂直起降机场中飞机的充电状态决定。记为 $\mathcal{W}_{v,v'}^t = \{t' \in \mathcal{T} \mid t' + o_{v,v'}^{t'} = t\}$ ，约束条件为:

$$z_{v,u}^t = z_{v,u}^{t-1} + \sum_{v' \in \mathcal{V} - \{v\}} \left(\sum_{i' \in \mathcal{W}_{v,v'}^t} y_{v',v,u+\mu_{v',v}^{t'},v}^{t'} - y_{v,v',u}^t \right) + \sum_{u'=0}^{u-1} x_{v,u',u}^t \left(t - \omega - \sum_{u''=u'+1}^u w_{u''} \right) - \sum_{u''=u+1}^U x_{v,u,u''}^t, \forall v, u, t. \quad (9)$$

System Stationarity. Ensure that the aircraft scheduling and charging policies can be repeated everyday. Previous studies on human mobility (Jiang et al., 2016) and travel surveys (US Department of Transportation, 2022) indicate consistent daily travel demand. Thus, by default, the day is treated as a closed system. However, in more dynamic areas without clear patterns, this constraint can be lifted. We can then adjust the optimization problem using model predictive control (MPC) (Garcia et al., 1989) to handle future demand prediction uncertainties.

系统平稳性。确保飞机调度和充电策略可以每天重复。先前关于人类出行的研究 (Jiang 等人, 2016 年) 和出行调查 (美国运输部, 2022 年) 表明每日出行需求具有一致性。因此, 默认情况下, 将一天视为一个封闭系统。然而, 在没有明确模式的更动态区域, 可以解除此约束。然后我们可以使用模型预测控制 (MPC)(Garcia 等人, 1989 年) 来调整优化问题, 以处理未来需求预测的不确定性。

$$x_{v,u,u'}^0 = x_{v,u,u'}^T, y_{v,v',u}^0 = y_{v,v',u}^T, z_{v,u}^0 = z_{v,u}^T, \forall v, v', u, u'. \quad (10)$$

3.5. Optimization problem

3.5. 优化问题

The optimization problem is:

优化问题如下:

$$\min \sum_{v,v',t} \left(q_{v,v'}^t - \sum_u y_{v,v',u}^t \right) \cdot \sum_{v,v',t} t_{v,v'}^t. \quad (11)$$

$$\text{s.t. } y_{v,v',0}^t = 0, \forall v, v', t, \quad (12)$$

$$q_{v,v'}^t - \sum_u y_{v,v',u}^t \geq 0, t_{v,v'}^t \geq 0, \forall v, v', t, \quad (13)$$

$$t_{v,v'}^t \geq q_{v,v'}^t - \sum_u y_{v,v',u}^t, \forall v, v', t, \quad (14)$$

$$\sum_{v,u,u'} x_{v,u,u'}^0 + \sum_{v,u,u'} x_{v,u,u'}^T + \sum_{v,u} z_{v,u}^0 = |\mathcal{A}|, \forall v, v', \quad (15)$$

$$z_{v,u}^t = z_{v,u}^{t-1} + \sum_{v' \in \mathcal{V} - \{v\}} \left(\sum_{i' \in \mathcal{W}_{v,v'}^t} y_{v',v,u+\mu_{v',v}^{t'},v}^{t'} - y_{v,v',u}^t \right)$$

$$+ \sum_{u'=0}^{u-1} x_{v,u',u}^t \left(t - \sum_{u''=u'+1}^u w_{u''} \right) - \sum_{u'''=u+1}^U x_{v,u,u'''}^t, \forall v, u, t, \quad (16)$$

$$x_{v,u,u'}^0 = x_{v,u,u'}^T, y_{v,v',u}^0 = y_{v,v',u}^T, z_{v,u}^0 = z_{v,u}^T, \forall v, v', u, u', \quad (17)$$

where the objective function of the optimization problem is to minimize the total number of unserved requests. Constraint (12) ensures aircraft of which the SoC is at reserve level cannot fly. Constraint (13)-(14) ensures that only non-negative unserved requests are feasible. Constraint (15) states the number of aircraft in the system equals the fleet size. Constraint (16) describes the dynamics of the number of aircraft at various vertiports by modeling the dynamics of the operations. Constraint (17) ensures the stationarity of the state of the system. The solution of $x_{v,u,u'}^t$ gives the charging time window and energy requirements. The solutions for $x_{v,u,u'}^0$, $y_{v,v',u}^0$, and $z_{v,u}^0$ determine the initial allocation of aircraft to vertiports.

其中，优化问题的目标函数是使未得到服务的请求总数最小化。约束条件 (12) 确保荷电状态处于储备水平的飞机不能飞行。约束条件 (13)-(14) 确保只有非负的未得到服务的请求是可行的。约束条件 (15) 表明系统中的飞机数量等于机队规模。约束条件 (16) 通过对运营动态进行建模来描述各个垂直机场的飞机数量动态。约束条件 (17) 确保系统状态的平稳性。 $x_{v,u,u'}^t$ 的解给出充电时间窗口和能量需求。 $x_{v,u,u'}^0$ 、 $y_{v,v',u}^0$ 和 $z_{v,u}^0$ 的解确定飞机到垂直机场的初始分配。

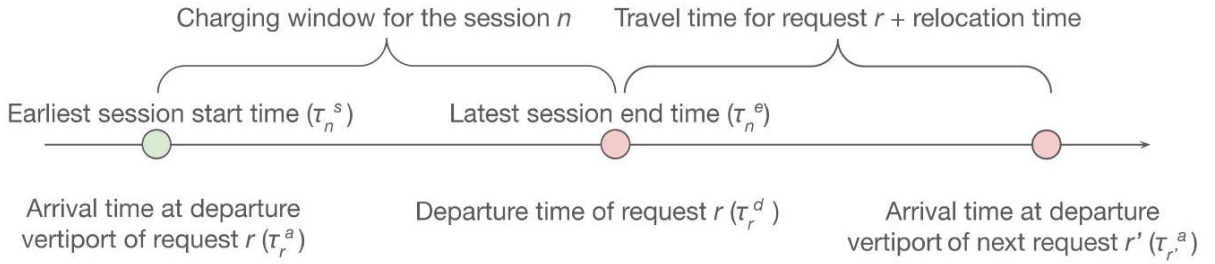


Fig. 2. Illustrative example of a charging window.

图 2. 充电窗口的示例。

4. Charging coordination considering power system dynamics

4. 考虑电力系统动态特性的充电协调

In this section, we model the interaction between the UAM system and the power grid. We propose a joint model to capture the coupling between these two systems. Using the charging time window and energy requirements derived from the optimal UAM fleet dispatch, we calculate the optimal charging schedule while considering power grid dynamics. Our goal is to maximize power system welfare through smart charging. We begin by formulating the problem in a deterministic setting and then extend it to incorporate stochastic factors.

在本节中，我们对城市空中交通 (UAM) 系统与电网之间的相互作用进行建模。我们提出一个联合模型来捕捉这两个系统之间的耦合关系。利用从最优 UAM 机队调度得出的充电时间窗口和能量需求，我们在考虑电网动态特性的同时计算最优充电计划。我们的目标是通过智能充电使电力系统福利最大化。我们首先在确定性环境中阐述问题，然后将其扩展以纳入随机因素。

4.1. System model

4.1. 系统模型

The system model includes both the UAM system and the power system. In the deterministic formulation, we assume that charging demand and the charging time window are known with certainty.

系统模型包括城市空中交通 (UAM) 系统和电力系统。在确定性公式中，我们假设充电需求和充电时间窗口是确定已知的。

UAM System. At the vertiport charging station, each charging session n is characterized by its earliest and latest possible charging times, denoted by τ_n^s and τ_n^e , respectively. Here, τ_n^s is no earlier than the time τ_r^a when the aircraft completes its previous travel request and repositions to the origin vertiport for its next travel request, while τ_n^e no later than the time τ_r^d when the aircraft departs from the vertiport. An illustrative example of the charging window is shown in Fig. 2. The charging demand for session n is denoted by d_n . The parameters τ_n^s , τ_n^e , and d_n are obtained from the fleet dispatch. The binary parameter s_n^t indicates the available charging time slots as follows:

城市空中交通 (UAM) 系统。在垂直机场充电站，每个充电会话 n 的特征是其最早和最晚可能充电时间，分别用 τ_n^s 和 τ_n^e 表示。这里， τ_n^s 不早于飞机完成其先前行程请求并重新定位到起始垂直机场以进行下一个行程请求的时间 τ_r^a ，而 τ_n^e 不晚于飞机从垂直机场起飞的时间 τ_r^d 。充电窗口的一个示例见图 2。会话 n 的充电需求用 d_n 表示。参数 τ_n^s , τ_n^e 和 d_n 从机队调度中获得。二元参数 s_n^t 指示可用充电时隙如下：

$$s_n^t = \begin{cases} 1, & \text{if } \tau_n^s \leq t \leq \tau_n^e, \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

Within the charging window, the operator must decide on the charging power for session n at time slot t as p_n^t , which is bounded by P^+ . The binary parameter $\phi_{n,m}$ indicates whether session n takes place at charging station m . The total charging capacity of station m is denoted by C_m^+ , meaning that the combined power of multiple concurrent charging sessions at the same station should not exceed C_m^+ .

在充电窗口内，运营商必须确定在时隙 t 时会话 n 的充电功率为 p_n^t ，其受限于 P^+ 。二元参数 $\phi_{n,m}$ 指示会话 n 是否在充电站 m 进行。充电站 m 的总充电容量用 C_m^+ 表示，这意味着同一充电站多个并发充电会话的组合功率不应超过 C_m^+ 。

Power System. The power system consists of buses and transmission lines as its fundamental components. Some buses function as generators, while others serve as loads. Buses that are neither generators nor loads act as interconnections. Generators produce power and deliver it to the network, while loads consume

power from the network. The operator determines the energy generation at bus i during time slot t as g_i^t , which is constrained by the minimum and maximum generation limits G_i^- and G_i^+ . The difference in generation between two consecutive time slots is restricted by the ramp-down and ramp-up limits, denoted by R_i^- and R_i^+ . The generated power supplies the charging load of vertiports (with the binary indicator $\eta_{n,i}$ showing whether session n is a load for bus i) as well as the external load ℓ_i^t of bus i at time slot t . The voltage angle at bus i in time slot t is represented by δ_i^t , and the reactance of the transmission line between buses i and j is $b_{i,j}$. The voltage angle and reactance together determine the power flow between buses i and j , denoted by $f_{i,j}$, which is constrained by the transmission line's power flow limits, $F_{i,j}^-$ and $F_{i,j}^+$.

电力系统。电力系统由母线和输电线路作为其基本组件组成。一些母线用作发电机，而其他母线用作负载。既不是发电机也不是负载的母线用作互连。发电机发电并将其输送到网络，而负载从网络消耗电力。运营商确定在时隙 t 时母线 i 的发电量为 g_i^t ，其受最小和最大发电限制 G_i^- 和 G_i^+ 的约束。两个连续时隙之间的发电量差异受降温和升温限制的约束，分别用 R_i^- 和 R_i^+ 表示。所发电力为垂直机场的充电负载供电（二元指示器 $\eta_{n,i}$ 显示会话 n 是否是母线 i ）的负载以及在时隙 t 时母线 i 的外部负载 ℓ_i^t ）。时隙 t 时母线 i 的电压角用 δ_i^t 表示，母线 i 和 j 之间输电线路的电抗为 $b_{i,j}$ 。电压角和电抗共同决定母线 i 和 j 之间的潮流，用 $f_{i,j}$ 表示，其受输电线路潮流限制 $F_{i,j}^-$ 和 $F_{i,j}^+$ 的约束。

4.2. Decision variable

4.2. 决策变量

We introduce variables at both the vertiport level and the power grid level. At the vertiport charging station level, the continuous decision variable is p_n^t , representing the charging power for session n at time slot t , and is bounded by P^+ . At the power grid level, the decision variables include the continuous variables for energy generation at bus i and time slot t as g_i^t , power flow between bus i and bus j at time slot t as $f_{i,j}^t$, and the voltage angle at bus i and time slot t as $\delta_{i,j}^t$.

我们在垂直机场层面和电网层面引入变量。在垂直机场充电站层面，连续决策变量为 p_n^t ，表示在时隙 t 时会话 n 的充电功率，且受限于 P^+ 。在电网层面，决策变量包括在母线 i 及时隙 t 时的发电连续变量 g_i^t 、在时隙 t 时母线 i 与母线 j 之间的潮流 $f_{i,j}^t$ ，以及在母线 i 及时隙 t 时的电压角度 $\delta_{i,j}^t$ 。

4.3. Objective

4.3. 目标

The objective is to maximize power system welfare. Denote $\Psi(\cdot)$ as the generation cost function and $\Theta_i(g_i^t)$ as the benefit the system can obtain by using the generated power g_i^t at bus i , we have:

目标是使电力系统福利最大化。将系统在母线 i 处使用发电量 g_i^t 所能获得的收益记为 $\Theta_i(g_i^t)$ ，我们有：

$$\max \sum_t \sum_i \Theta_i(g_i^t) - \sum_t \sum_i \Psi_i(g_i^t) \quad (19)$$

4.4. Constraints

4.4. 约束条件

On the charging station level, there are charging demand constraints, charging window constraints as well as charging station capacity constraints.

在充电站层面，存在充电需求约束、充电时段约束以及充电站容量约束。

Energy Demand Constraints. Ensure the charging demand for each session should be satisfied.

能源需求约束。确保满足每个时段的充电需求。

$$\sum_t p_n^t = d_n, \forall n \in \mathcal{N}, \quad (20)$$

Charging Window Constraints. Ensure the power can be positive only during the charging window of the session. When the time t is out of the charging session n , i.e., $s_n^t = 0$, the constraint becomes $p_n^t = 0$. When the time t is in the charging session n , i.e., $s_n^t = 1$, the constraint becomes $0 \leq p_n^t \leq P^+$.

充电时段约束。确保功率仅在时段的充电窗口内为正。当时间 t 不在时段 n 的充电窗口内，即 $s_n^t = 0$ 时，约束变为 $p_n^t = 0$ 。当时间 t 在时段 n 的充电窗口内，即 $s_n^t = 1$ 时，约束变为 $0 \leq p_n^t \leq P^+$ 。

$$0 \leq p_n^t \leq s_n^t P^+, \forall n \in \mathcal{N}, t \in \mathcal{T}, \quad (21)$$

Station Capacity Constraints. Ensure the maximal charging capacity of each vertiport is not exceeded. $\phi_{n,m} p_n^t$ is the total power at station m in time t . By enforcing $0 \leq \sum_n \phi_{n,m} p_n^t \leq C_m^+$, we ensure the total power at every station in every time slot does not its capacity C_m^+ .

充电站容量约束。确保不超过每个垂直机场的最大充电容量。 $\phi_{n,m} p_n^t$ 是时段 t 内充电站 m 的总功率。通过实施 $0 \leq \sum_n \phi_{n,m} p_n^t \leq C_m^+$ ，我们确保每个时段每个充电站的总功率不超过其容量 C_m^+ 。

$$0 \leq \sum_n \phi_{n,m} p_n^t \leq C_m^+, \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (22)$$

On the grid level, we consider power flow dynamics, which need to consider (1) how the power injections are distributed along the power lines, and (2) how to ensure that the power flows do not exceed the capacity of the lines. To model flow dynamics, both the higher-fidelity AC model (Glover et al., 2017) and the simplified linear DC model (Stott et al., 2009) are commonly used. The AC model accurately captures the physical behavior of the network but becomes computationally prohibitive for large-scale optimization. On the other hand, the DC model, due to its linearity, is well-suited for large-scale problems. The linearity of the DC model relies on several key assumptions: (1) voltage magnitudes are constant, (2) reactive power and line resistances are neglected, and (3) phase angle differences between buses are assumed to be small. While these assumptions improve computational efficiency, they also limit the model's ability to handle voltage constraints

in distribution networks (Hogan, 1993). Despite this, the DC model remains widely used in transmission-level application (Overbye et al., 2004) and is appropriate for high-level analysis of control policies (Rossi et al., 2019). In this paper, we focus on the transmission-level application.

在电网层面，我们考虑潮流动态，这需要考虑 (1) 功率注入如何沿电力线路分布，以及 (2) 如何确保潮流不超过线路容量。为了对潮流动态进行建模，通常使用高保真交流模型 (Glover 等人, 2017 年) 和简化的线性直流模型 (Stott 等人, 2009 年)。交流模型准确地捕捉了网络的物理行为，但对于大规模优化来说计算成本过高。另一方面，直流模型由于其线性，非常适合大规模问题。直流模型的线性依赖于几个关键假设: (1) 电压幅值恒定，(2) 无功功率和线路电阻被忽略，以及 (3) 母线之间的相角差被假定为很小。虽然这些假设提高了计算效率，但它们也限制了模型处理配电网中电压约束的能力 (Hogan, 1993 年)。尽管如此，直流模型在输电层面的应用中仍然被广泛使用 (Overbye 等人, 2004 年)，并且适用于控制策略的高级分析 (Rossi 等人, 2019 年)。在本文中，我们专注于输电层面的应用。

System Balance Constraints. Ensure the difference between outflow and inflow equals the net load generation minus load.

系统平衡约束。确保流出量和流入量之间的差值等于净负荷发电量减去负荷。

$$g_i^t - \left(\ell_i^t + \sum_n \eta_{n,i} p_n^t \right) = \sum_k f_{i,k} - \sum_j f_{j,i}, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (23)$$

$$f_{i,j}^t = b_{i,j} (\delta_i^t - \delta_j^t), \quad \forall i, j \in \mathcal{I}, t \in \mathcal{T}. \quad (24)$$

$$-2\pi \leq \delta_i^t \leq 2\pi, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (25)$$

Generation Capacity Constraints. Ensure the generation does not exceed generator capacity lower bound G_i^- and upper bound G_i^+ .

发电容量约束。确保发电量不超过发电机容量下限 G_i^- 和上限 G_i^+ 。

$$G_i^- \leq g_i^t \leq G_i^+, \quad \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (26)$$

Flow Capacity Constraints. Ensure the flow does not exceed transmission line capacity. The $F_{i,j}^-$ and $F_{i,j}^+$ has same absolute value but different sign, i.e, $F_{i,j}^- = -F_{i,j}^+$, which ensures for bi-directional flow does not exceed $F_{i,j}^+$.

潮流容量约束。确保潮流不超过输电线路容量。 $F_{i,j}^-$ 和 $F_{i,j}^+$ 具有相同的绝对值但符号不同，即 $F_{i,j}^- = -F_{i,j}^+$ ，这确保双向潮流不超过 $F_{i,j}^+$ 。

$$F_{i,j}^- \leq f_{i,j}^t \leq F_{i,j}^+, \quad \forall i, j \in \mathcal{I}, t \in \mathcal{T}. \quad (27)$$

Ramping Capacity Constraints. Ensure the generation power gap between consecutive time intervals does not exceed the lower limit R_i^- and upper limit R_i^+ .

爬坡容量约束。确保连续时间间隔之间的发电功率差距不超过下限 R_i^- 和上限 R_i^+ 。

$$R_i^- \leq g_i^t - g_i^{t-1} \leq R_i^+, \forall i \in \mathcal{I}, t \in \mathcal{T}. \quad (28)$$

4.5. Optimization problem

4.5. 优化问题

Together, we have the optimization problem as:

我们共同拥有的优化问题如下:

$$\max \sum_t \sum_i \Theta_i(g_i^t) - \sum_t \sum_i \Psi_i(g_i^t) \quad (29)$$

$$\text{s.t.} \quad \sum_t p_n^t = d_n, \forall n \in \mathcal{N}, \quad (30)$$

$$0 \leq p_n^t \leq s_n^t P^+, \forall n \in \mathcal{N}, t \in \mathcal{T}, \quad (31)$$

$$0 \leq \sum_n \phi_{n,m} p_n^t \leq C_m^+, \forall m \in \mathcal{M}, t \in \mathcal{T}, \quad (32)$$

$$g_i^t - \left(\ell_i^t + \sum_n \eta_{n,i} p_n^t \right) = \sum_k f_{i,k} - \sum_j f_{j,i}, \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (33)$$

$$f_{i,j}^t = b_{i,j} (\delta_i^t - \delta_j^t), \forall i, j \in \mathcal{I}, t \in \mathcal{T} \quad (34)$$

$$-2\pi \leq \delta_i^t \leq 2\pi, \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (35)$$

$$F_{i,j}^- \leq f_{i,j}^t \leq F_{i,j}^+, \forall i, j \in \mathcal{I}, t \in \mathcal{T}. \quad (36)$$

$$G_i^- \leq g_i^t \leq G_i^+, \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (37)$$

$$R_i^- \leq g_i^t - g_i^{t-1} \leq R_i^+, \forall i \in \mathcal{I}, t \in \mathcal{T}, \quad (38)$$

where Eq. (39) maximizes the power system welfare of the system. Constraint (30) describes the charging requirement. Constraint (31) characterizes the maximal charging power of each session in each vertiport. Constraint (32) characterizes the maximal charging capacity of each vertiport. Constraints (33)-(35) denotes system balance. Constraints (36) describe the line capacity constraints. Constraint (37) denotes the generation capacity. Constraints (38) characterize the ramping constraints.

其中，式 (39) 使系统的电力系统福利最大化。约束条件 (30) 描述了充电需求。约束条件 (31) 表征了每个垂直机场中每个会话的最大充电功率。约束条件 (32) 表征了每个垂直机场的最大充电容量。约束条件 (33)-(35) 表示系统平衡。约束条件 (36) 描述了线路容量约束。约束条件 (37) 表示发电容量。约束条件 (38) 表征了爬坡约束。

5. Charging coordination considering uncertainty

5. 考虑不确定性的充电协调

The deterministic formulation focuses on cases where the aircraft's arrival time and demand are accurately estimated. However, in reality, the arrival time τ_n^s and charging demand d_n for each aircraft are stochastic. The uncertainty in the system affects the flexibility that smart charging can provide. For instance, a narrowed time window reduces the flexibility for adjusting the charging schedule. Similarly, if an aircraft consumes more energy than expected, the demand may be higher than anticipated.

确定性公式侧重于飞机到达时间和需求能够被准确估计的情况。然而，在现实中，每架飞机的到达时间 τ_n^s 和充电需求 d_n 都是随机的。系统中的不确定性会影响智能充电所能提供的灵活性。例如，时间窗口变窄会降低调整充电计划的灵活性。同样，如果一架飞机消耗的能量比预期多，需求可能会高于预期。

5.1. Variables and parameters

5.1. 变量和参数

The stochastic formulation shares the same variables and parameters as the deterministic formulation. The difference between the deterministic and stochastic formulations lies in the randomness of the variables. For each session n , its demand d_n and start time τ_n^s are stochastic and are sampled from a joint distribution Ω_n , i.e., $d_n, \tau_n^s \sim \Omega_n$.

随机公式与确定性公式共享相同的变量和参数。确定性公式和随机公式之间的区别在于变量的随机性。对于每个会话 n ，其需求 d_n 和开始时间 τ_n^s 是随机的，并且从联合分布 Ω_n 中采样，即 $d_n, \tau_n^s \sim \Omega_n$ 。

We assume the functions describe the probability of finite discrete realizations. Suppose we have $k \in \mathcal{K}$ realizations. A realization of the possible pattern is noted by the parameters $d_{n,k}$ and $s_{n,k}^t$ associated with the probability $\pi_{n,k}$. A binary variable $h_{n,k}$ is introduced for each pattern realization.

我们假设这些函数描述了有限离散实现的概率。假设我们有 $k \in \mathcal{K}$ 个实现。可能模式的一个实现由与概率 $\pi_{n,k}$ 相关联的参数 $d_{n,k}$ 和 $s_{n,k}^t$ 表示。为每个模式实现引入一个二元变量 $h_{n,k}$ 。

5.2. Objective

5.2. 目标

The objective is still to maximize the total power system welfare. Denote $\Psi(\cdot)$ as the generation cost function and $\Theta_i(g_i^t)$ as the benefit the system can obtain by using the generated power g_i^t at bus i , we have:

目标仍然是使电力系统总福利最大化。将 $\Psi(\cdot)$ as the generation cost function and $\Theta_i(g_i^t)$ 表示为系统在母线 i 处使用发电功率 g_i^t 可获得的收益，我们有：

$$\max \sum_t \sum_i \Theta_i(g_i^t) - \sum_t \sum_i \Psi_i(g_i^t) \quad (39)$$

5.3. Constraints

5.3. 约束条件

The stochastic formulation retains the same constraints as the deterministic formulation, except for Constraints (30)-(31). In the stochastic formulation, the charging demand d_n and the earliest start time t_n^{str} for session n are treated as stochastic parameters influenced by aircraft operations. We reformulate Constraints (30) and (31) using a chance-constrained framework as follows:

随机公式保留了与确定性公式相同的约束条件，但不包括约束条件 (30)-(31)。在随机公式中，充电需求 d_n 和会话 n 的最早开始时间 t_n^{str} 被视为受飞机运行影响的随机参数。我们使用机会约束框架重新制定约束条件 (30) 和 (31) 如下：

$$\Pr \left(\begin{array}{l} d_n^- \leq \sum_t p_n^t - d_n \leq d_n^+ \\ 0 \leq p_n^t \leq s_n^t P^+ \end{array}, \forall t \in \mathcal{T} \right) \geq 1 - \epsilon, \forall n \in \mathcal{N}, \quad (40)$$

where ϵ is a predefined confidence parameter for each session. This constraint ensures that the probability of failing to meet the charging requirements is below ϵ .

其中 ϵ 是每个会话的预定义置信参数。此约束确保无法满足充电要求的概率低于 ϵ 。

The distribution of stochastic parameters is hard to model by a continuous function. Also, these two parameters are correlated with each other. Here we assume that stochastic parameters have a finite discrete distribution, and then we can solve it by a mixed-integer programming method (Luedtke et al., 2010). We can divide the pattern realizations into two sets.

随机参数的分布很难用连续函数建模。此外，这两个参数相互关联。在这里，我们假设随机参数具有有限的离散分布，然后我们可以通过混合整数规划方法求解 (Luedtke 等人, 2010)。我们可以将模式实现分为两组。

Essential Requirements. The subset of possible pattern realizations with a probability over the confidence parameter is defined as:

基本要求。概率超过置信参数的可能模式实现子集定义为:

$$\mathcal{K}_n^+ = \{k \in \mathcal{K}_n, \pi_{n,k} > \epsilon\}, \quad (41)$$

The binary variable for the realization k should be $h_{n,k} = 0$ when $\pi_{n,k} > \epsilon$. Since if this realization is true, the probability will exceed the ϵ . For realization $k \in \mathcal{K}_n^+$, the Constraints are as original:

当 $\pi_{n,k} > \epsilon$ 时, 实现 k 的二元变量应为 $h_{n,k} = 0$ 。因为如果这个实现为真, 概率将超过 ϵ 。对于实现 $k \in \mathcal{K}_n^+$, 约束条件与原来相同:

$$d_n^- \leq \sum_t p_n^t - d_{n,k} \leq d_n^+, \quad \forall n \in \mathcal{N}, k \in \mathcal{K}_n^+, \quad (42)$$

$$0 \leq p_n^t \leq s_{n,k}^t P^+, \quad \forall n \in \mathcal{N}, k \in \mathcal{K}_n^+. \quad (43)$$

These constraints ensure all realizations with probability more than ϵ should be satisfied.

这些约束确保所有概率超过 ϵ 的实现都应得到满足。

Nonessential Requirements. The subset of possible pattern realizations with a probability less than or equal to the confidence parameter is defined as:

非基本要求。概率小于或等于置信参数的可能模式实现子集定义为:

$$\mathcal{K}_n^- = \{k \in \mathcal{K}_n, \pi_{n,k} \leq \epsilon\}. \quad (44)$$

For realization $k \in \mathcal{K}_n^-$, the Constraints can be reformulated as

对于实现 $k \in \mathcal{K}_n^-$, 约束条件可以重新制定为

$$\sum_t p_n^t - d_{n,k} - h_{n,k} \left(\sum_t P^+ - d_{n,k} \right) \leq d_n^+, \quad \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (45)$$

$$\sum_t p_n^t - d_{n,k} + h_{n,k} d_{n,k} \geq d_n^-, \quad \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (46)$$

$$0 \leq p_n^t \leq P^+ (s_{n,k}^t + h_{n,k}), \quad \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, t \in \mathcal{T}, \quad (47)$$

These constraints ensure all realizations with probability less or equal to ϵ can be selected to be compromised.

这些约束确保所有概率小于或等于 ϵ 的实现都可以被选择进行折衷。

When the binary variable $h_{n,k} = 0$, Constraints are the same as the deterministic cases:

当二元变量 $h_{n,k} = 0$ 时, 约束条件与确定性情况相同:

$$d_n^- \leq \sum_t p_n^t - d_{n,k} \leq d_n^+, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (48)$$

$$0 \leq p_n^t \leq s_{n,k}^t P^+, \forall n \in \mathcal{N}, t \in \mathcal{T}, k \in \mathcal{K}_n^- \quad (49)$$

When the binary variable $h_{n,k} = 1$, Constraints are relaxed as:

当二元变量 $h_{n,k} = 1$ 时，约束条件放宽为：

$$\sum_t p_n^t - d_{n,k} - \sum_t P^+ \leq d_n^+, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (50)$$

$$\sum_t p_n^t \geq d_n^-, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (51)$$

$$0 \leq p_n^t \leq P^+ (s_{n,k}^t + 1), \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, t \in \mathcal{T}, \quad (52)$$

Constraints (50) is guaranteed since $\sum_t p_n^t \leq \sum_t P^+ + d_{n,k} + d_n^+$ is always true. Since we have $d_n^- \leq 0$, we can guarantee Constraints (51) too. For Constraints (52), if $s_{n,k}^t = 0$, we have $0 \leq p_n^t \leq P^+$, which means even if the aircraft will not able to charge at time slot t , charging now is still a feasible solution in this optimization problem. If $s_{n,k}^t = 1$, we have $0 \leq p_n^t \leq 2P^+$, which means the aircraft can charge at time slot t . Together this constraint ensures that whether this time slot t is available for charging or not, a positive charging power solution is feasible.

约束条件 (50) 是有保证的，因为 $\sum_t p_n^t \leq \sum_t P^+ + d_{n,k} + d_n^+$ 总是为真。由于我们有 $d_n^- \leq 0$ ，我们也可以保证约束条件 (51)。对于约束条件 (52)，如果 $s_{n,k}^t = 0$ ，我们有 $0 \leq p_n^t \leq P^+$ ，这意味着即使飞机在时隙 t 无法充电，现在充电在这个优化问题中仍然是一个可行的解决方案。如果 $s_{n,k}^t = 1$ ，我们有 $0 \leq p_n^t \leq 2P^+$ ，这意味着飞机可以在时隙 t 充电。这个约束共同确保了无论这个时隙 t 是否可用于充电，正充电功率解决方案都是可行的。

Failure Probability Requirements. Ensures the summation of probability $\pi_{n,k}$ under all realizations for session n is less than ϵ . The subset of possible pattern realizations with a probability over the confidence parameter is defined as:

故障概率要求。确保会话 n 的所有实现下概率 $\pi_{n,k}$ 的总和小于 ϵ 。概率超过置信参数的可能模式实现子集定义为：

$$\sum_{k \in \mathcal{K}_n^+} (\pi_{n,k} h_{n,k}) \leq 0, \forall n \in \mathcal{N}, \quad (53)$$

$$h_{n,k} \in \{0, 1\}, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^+. \quad (54)$$

The subset of possible pattern realizations with a probability less or equal to the confidence parameter is defined as:

概率小于或等于置信参数的可能模式实现子集定义为：

$$\sum_{k \in \mathcal{K}_n^-} (\pi_{n,k} h_{n,k}) \leq \epsilon, \forall n \in \mathcal{N}, \quad (55)$$

$$h_{n,k} \in \{0, 1\}, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-. \quad (56)$$

5.4. Optimization problem

5.4. 优化问题

We have the optimization problem as

我们有如下优化问题

$$\max \sum_t \sum_i \Theta_i(g_i^t) - \sum_t \sum_i \Psi_i(g_i^t) \quad (57)$$

s.t. Constraints (32)-(38)

受限于约束条件 (32)-(38)

(58)

$$d_n^- \leq \sum_t p_n^t - d_{n,k} \leq d_n^+, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^+, \quad (59)$$

$$0 \leq p_n^t \leq s_{n,k}^t P^+, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^+. \quad (60)$$

$$\sum_{k \in \mathcal{K}_n^+} (\pi_{n,k} h_{n,k}) \leq 0, \forall n \in \mathcal{N}, \quad (61)$$

$$h_{n,k} \in \{0, 1\}, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^+. \quad (62)$$

$$\sum_t p_n^t - d_{n,k} - h_{n,k} \left(\sum_t P^+ - d_{n,k} \right) \leq d_n^+, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (63)$$

$$\sum_t p_n^t - d_{n,k} + h_{n,k} d_{n,k} \geq d_n^-, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (64)$$

$$0 \leq p_n^t \leq P^+ (s_{n,k}^t + h_{n,k}), \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, t \in \mathcal{T}, \quad (65)$$

$$\sum_{k \in \mathcal{K}_n^-} (\pi_{n,k} h_{n,k}) \leq \epsilon, \forall n \in \mathcal{N}, \quad (66)$$

$$h_{n,k} \in \{0, 1\}, \forall n \in \mathcal{N}, k \in \mathcal{K}_n^-, \quad (67)$$

Table 1

表 1

Aircraft configuration.

飞机配置。

Name	Charging power [kW]	Battery capacity [kWh]	Initial SoC [%]	Reserve SoC [%]	SoC levels	Energy rate [kWh/mile]	Passenger capacity
Value	250	160	70	20	8	1	1

Table 2

表 2

Pattern realizations and probabilities.

模式实现与概率。

Pattern realizations	1	2	3	4	5	6	7	8	9
Arrival time [min]	-10	-10	-10	+0	+0	+0	+10	+10	+10
Energy demand [kWh]	-30	+0	+30	-30	+0	+30	-30	+0	+30
Probability	0.00020	0.0063	0.00020	0.029	0.93	0.029	0.00020	0.0063	0.00020

where Eq. (57) maximizes the power system welfare. Constraints (32)-(38) describe deterministic power system dynamics. Constraints (59)-(62) ensure the subset of possible pattern realizations with a probability over the confidence parameter should be satisfied. Constraints (63)-(67) ensure the summation of probability under all realizations for each session is less than the predetermined confidence level.

其中，式 (57) 使电力系统福利最大化。约束条件 (32)-(38) 描述了确定性电力系统动态。约束条件 (59)-(62) 确保应满足概率超过置信参数的可能模式实现子集。约束条件 (63)-(67) 确保每个会话的所有实现下的概率总和小于预定置信水平。

6. Numerical studies

6. 数值研究

We first introduce the dataset used and then present the UAM travel demand. Following this, we demonstrate the flexibility available in both the deterministic and stochastic settings.

我们首先介绍所使用的数据集，然后展示城市空中交通 (UAM) 的出行需求。在此之后，我们展示在确定性和随机性设置中可用的灵活性。

6.1. Data overview

6.1. 数据概述

We use Location-Based Service (LBS) data to model the mobility trajectory of users. The datasets, as described in Cuebiq (2023), cover Miami-Dade County in Florida and were collected over a two-month period from September to October 2022. The total number of users is approximately 100,000, and these users generated millions of records within the given region and timeframe. Each record consists of a user ID, a timestamp, a stay duration, and a pair of geographical coordinates.

我们使用基于位置的服务 (LBS) 数据来模拟用户的移动轨迹。如 Cuebiq(2023) 所述, 这些数据集覆盖佛罗里达州的迈阿密 - 戴德县, 是在 2022 年 9 月至 10 月的两个月期间收集的。用户总数约为 100,000, 这些用户在给定区域和时间范围内生成了数百万条记录。每条记录包括用户 ID、时间戳、停留时长和一对地理坐标。

For the power grid analysis, we utilize a synthetic 9-bus transmission system (Pypower, 2024) to evaluate the performance of our framework. This system includes power flow data for nine buses, three of which are generators. The dataset comprises bus data, generator data, branch data, and generator cost data. Each bus can function as a generator or not; if it does, the data includes the maximum and minimum power outputs. Each generator has an associated cost represented by a quadratic function. For each branch between two buses, the dataset provides its reactance, as well as the lower and upper power capacity. Each bus has a load that must be satisfied, which is scaled from the data provided by US Energy Information Administration (2024). This data offers hourly load information for a typical day in Florida.

对于电网分析, 我们利用一个合成的 9 节点输电系统 (Pypower, 2024) 来评估我们框架的性能。该系统包括九个节点的潮流数据, 其中三个是发电机。数据集包括节点数据、发电机数据、支路数据和发电机成本数据。每个节点可以作为发电机, 也可以不作为发电机; 如果作为发电机, 数据包括最大功率输出和最小功率输出。每个发电机都有一个由二次函数表示的相关成本。对于两个节点之间的每条支路, 数据集提供其电抗以及功率容量的下限和上限。每个节点都有一个必须满足的负荷, 该负荷是根据美国能源信息管理局 (2024) 提供的数据进行缩放的。此数据提供了佛罗里达州典型一天的每小时负荷信息。

6.2. Experiments setting

6.2. 实验设置

Travel Demand. Denote the saved travel time compared to traditional cars as Δt and the travel distance as d , we define $L = [\Delta t, d]$ as the factor vector to determine most possible 70,000 UAM trips. These trips satisfied $\Delta t > 30$ min and $d > 30$ miles.

出行需求。将与传统汽车相比节省的出行时间记为 Δt , 出行距离记为 d , 我们将 $L = [\Delta t, d]$ 定义为确定最可能的 70000 次城市空中出行 (UAM) 行程的因子向量。这些行程满足 $\Delta t > 30$ min 和 $d > 30$ 英里。

Fleet Dispatch. Miami-Dade County is divided into hexagons with an edge length of 10 km, each containing a vertiport, resulting in a total of 27 stations. We assume the fleet size is 1500 and the flexible time window is set to be 20 min. The charging demand within the county is treated as the load for bus 1. The aircraft configuration is shown in Table 1. In the numerical study, we assume the energy consumption per mile is a constant that stays the same across the day.

机队调度。迈阿密 - 戴德县被划分为边长为 10 km 的六边形，每个六边形包含一个垂直起降机场，总共 27 个站点。我们假设机队规模为 1500，灵活时间窗口设置为 20 分钟。县内的充电需求被视为公交车 1 的负荷。飞机配置如表 1 所示。在数值研究中，我们假设每英里的能耗是一个常数，全天保持不变。

Power Grid. The generation capacity, ramping limits, line capacity, and reactance are derived from the 9-bus system data. In this case study, we set the generation function as a quadratic function, i.e., $\Psi(x) = \xi_1 x^2 + \xi_2 x + \xi_3$, where the ξ_1, ξ_2, ξ_3 is obtained from 9-bus system data. The benefit function is assumed to be $\Theta(x) = \sigma x$, where σ can be regarded as the amortized benefit the users can get from using one unit of electricity. Here we set it as the 0.1 S/kWh, which is at the same magnitude as the electricity price.

电网。发电容量、爬坡限制、线路容量和电抗均来自 9 节点系统数据。在本案例研究中，我们将发电函数设置为二次函数，即 $\Psi(x) = \xi_1 x^2 + \xi_2 x + \xi_3$ ，其中 ξ_1, ξ_2, ξ_3 从 9 节点系统数据中获取。效益函数假设为 $\Theta(x) = \sigma x$ ，其中 σ 可视为用户使用一单位电力可获得的分摊效益。在此我们将其设置为 0.1 S/kWh，其大小与电价相同。

Uncertainty Modeling. For stochastic formulation, the confidence level is set as 0.05. The pattern realizations and probability for each session are listed in Table 2. In the table, we show the difference between the sampled arrival time (energy demand) and the mean arrival time (energy demand). The probability is sampled from a joint normal distribution with a mean value as the arrival time and energy demand in the deterministic formulation.

不确定性建模。对于随机公式，置信水平设置为 0.05。每个时段的模式实现和概率列于表 2 中。在表中，我们展示了采样到达时间 (能源需求) 与平均到达时间 (能源需求) 之间的差异。概率是从联合正态分布中采样得到的，其均值为确定性公式中的到达时间和能源需求。

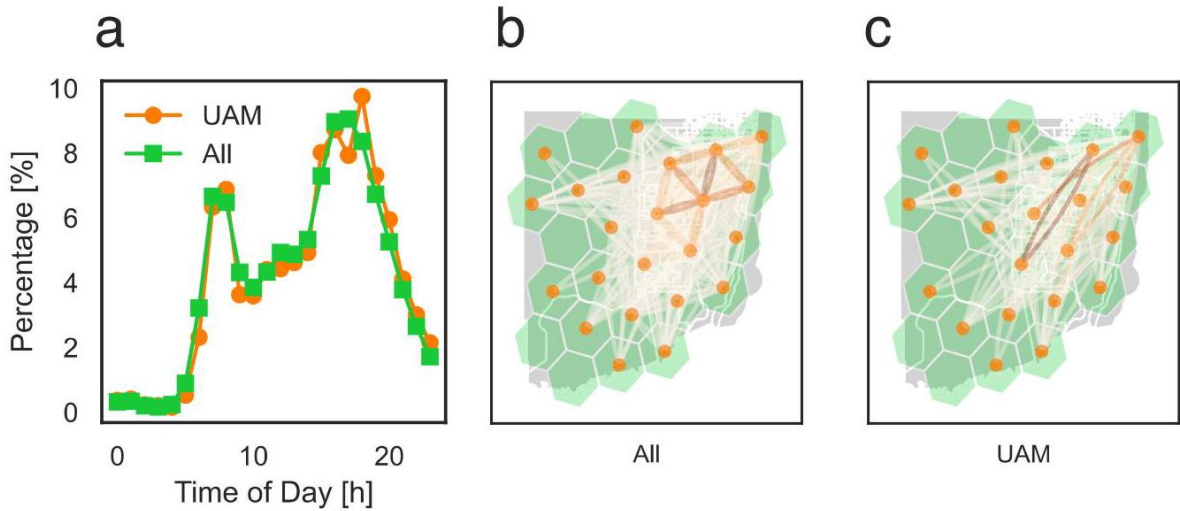


Fig. 3. Comparison of all travel demand and the UAM demand. (a) Temporal distribution of the departure time of all trips and UAM trips. (b) Geographical distribution of all trips' OD. (c) Geographical distribution of UAM trips' OD.

图 3. 所有出行需求与城市空中出行 (UAM) 需求的比较。(a) 所有行程和 UAM 行程出发时间的时间分布。(b) 所有行程起讫点 (OD) 的地理分布。(c) UAM 行程起讫点 (OD) 的地理分布。

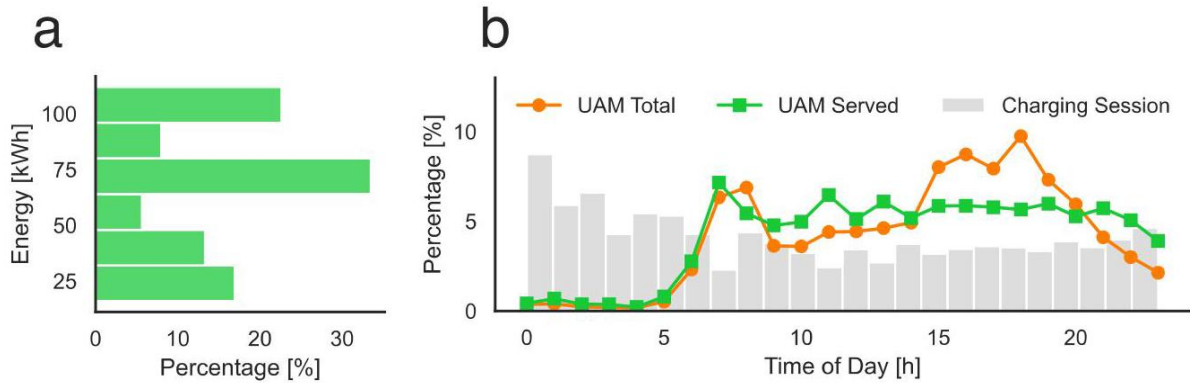


Fig. 4. Charging session overview. (a) Histogram of charging session energy. (b) Histogram of charging session start time, UAM trip departure time distribution, and served UAM trip departure time distribution.

图 4. 充电时段概述。(a) 充电时段能量直方图。(b) 充电时段开始时间、UAM 行程出发时间分布和已服务 UAM 行程出发时间分布的直方图。

6.3.UAM charging demand

6.3. 城市空中交通充电需求

We analyze the differences between the city-wide mobility pattern and the UAM travel demand. Our comparison includes the departure timing distribution as well as maps of the origin and destination matrix for both general travel demand and UAM travel demand. Fig. 3(a) illustrates that the departure timing for UAM and general travel demand have similar distributions. Fig. 3 (b-c) highlights the differences between general travel demand and UAM travel demand. The gray area represents the base map of Miami-Dade County. The green hexagons indicate areas, each containing one vertiport, represented by orange nodes. Links between vertiports signify travel demand between them, with the color of the edges indicating the flow of travel demand-the darker the color, the greater the flow. Our analysis reveals that while general travel demand covers the entirety of Miami-Dade County, UAM travel demand has more long trips between the northern and southern parts of the county.

我们分析了全市范围内的出行模式与城市空中交通出行需求之间的差异。我们的比较包括出发时间分布以及一般出行需求和城市空中交通出行需求的起讫点矩阵地图。图 3(a) 表明，城市空中交通和一般出行需求的出发时间分布相似。图 3(b - c) 突出了一般出行需求和城市空中交通出行需求之间的差异。灰色区域代表迈阿密 - 戴德县的基础地图。绿色六边形表示区域，每个区域包含一个由橙色节点代表的垂直机场。垂直机场之间的连线表示它们之间的出行需求，边的颜色表示出行需求的流量——颜色越深，流量越大。我们的分析表明，虽然一般出行需求覆盖了迈阿密 - 戴德县的整个区域，但城市空中交通出行需求在该县北部和南部之间有更多的长途行程。

Then we further show the charging session overview after optimal dispatch. Fig. 4(a) shows the distribution of energy demand of charging sessions. Most charging sessions have to satisfy a charging demand of around 75kWh. Fig. 4(b) shows the distribution of charging start time, the travel demand distribution, and served travel demand distribution. Most charging session starts at late night and early morning. This is because the travel demand is low late night and early morning, allowing for time for charging. During the day, the charging demand does not increase with more travel demand. This is because all aircraft are active in serving travel demand. From 0:00 to 6:00, travel demand is low, allowing most aircraft to charge, resulting in numerous charging sessions. During the 6:00 to 9:00 period, morning travel peaks occur, with many aircraft less available for charging due to active service. With full charges between 0:00-6:00, the fleet can meet trip demands, showing a served demand peak. From 9:00 to 15:00, demand drops, and so as the served demand. The 15:00-19:00 period has a second demand peak, but served demand does not rise due to limited fleet size and the morning peak depletes energy; aircraft charge only when SoC is critically low. From 19:00-24:00, demand decreases, allowing more charging time, and increasing charging sessions. This framework also provides important insights into optimizing capacity, the quantity of charging piles, and the allocation of parking spaces to reduce unmet travel demand. Specifically, the number of simultaneous sessions varies from 1 to 192, indicating significant differences in vertiport size requirements across different locations.

然后我们进一步展示了最优调度后的充电会话概述。图 4(a) 显示了充电会话的能量需求分布。大多数充电会话必须满足大约 75kWh 的充电需求。图 4(b) 显示了充电开始时间分布、出行需求分布和已服务出行需求分布。大多数充电会话在深夜和凌晨开始。这是因为深夜和凌晨出行需求较低，有时间进行充电。在白天，充电需求不会随着更多的出行需求而增加。这是因为所有飞机都在积极服务出行需求。从 0:00 到 6:00，出行需求较低，大多数飞机可以充电，导致大量充电会话。在 6:00 到 9:00 期间，出现早高峰出行，由于飞机在积极服务，许多飞机无法充电。在 0:00 - 6:00 之间充满电后，机队可以满足出行需求，显示出已服务需求高峰。从 9:00 到 15:00，需求下降，已服务需求也随之下降。15:00 - 19:00 期间有第二个需求高峰，但由于机队规模有限且早高峰消耗了能量，已服务需求没有上升；飞机仅在电池荷电状态极低时才充电。从 19:00 - 24:00，需求下降，允许更多充电时间，充电会话增加。该框架还为优化容量、充电桩数量和停车位分配以减少未满足的出行需求提供了重要见解。具体而言，同时会话的数量从 1 到 192 不等，表明不同地点对垂直机场规模的要求存在显著差异。

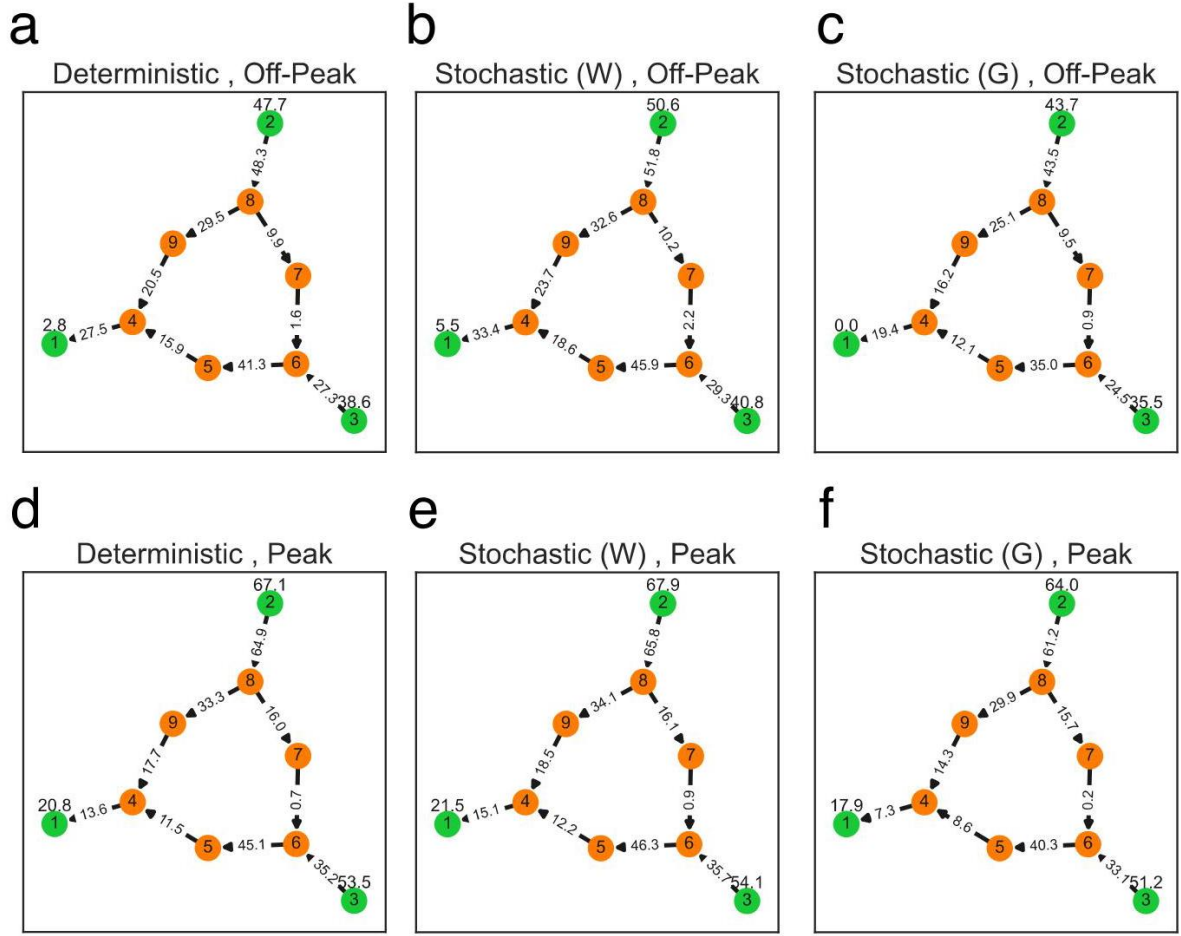


Fig. 5. Power flow to line capacity and power generation to generator capacity under Deterministic and Stochastic (power system welfare) and Stochastic (Generation). (a-c) Peak hour case. (d-f) Off-peak hour case.

图 5. 确定性和随机 (电力系统福利) 以及随机 (发电) 情况下线路容量的潮流和发电机容量的发电量。(a - c) 高峰时段情况。(d - f) 非高峰时段情况。

6.4. Power grid flow

6.4. 电网潮流

We analyze the power flow on transmission lines and the generation power at each generator for both deterministic and stochastic scenarios. Fig. 5 shows power flow in the system. The green nodes represent buses with generators, while the red nodes represent buses without generators. Arrows between two links indicate a flow on the transmission line, and the number on each arrow denotes the percentage of flow relative to the transmission line capacity. Similarly, the number on each node indicates the generation power relative to the generator's capacity. We define the peak hour as the time slot when the external load reaches its maximum and the off-peak hour as the time slot when the external load is at its minimum. Fig. 5 (a-c) display the off-peak hour power flow. In these scenarios, the stochastic scenario focusing on generation exhibits

a lower generation than the deterministic scenario, while the stochastic scenario focusing on power system welfare shows a higher generation than the deterministic scenario. Fig. 5 (d-f) show the peak hour power flow. Similarly, during peak hours, the deterministic scenario has a generation between stochastic scenarios focusing on power system welfare and a scenario focusing on generation.

我们分析了确定性和随机场景下输电线路上的潮流以及每个发电机的发电功率。图 5 展示了系统中的潮流。绿色节点表示有发电机的母线，而红色节点表示没有发电机的母线。两个链路之间的箭头表示输电线路上的潮流，每个箭头上的数字表示相对于输电线路容量的潮流百分比。同样，每个节点上的数字表示相对于发电机容量的发电功率。我们将高峰时段定义为外部负载达到最大值的时段，将非高峰时段定义为外部负载处于最小值的时段。图 5(a - c) 展示了非高峰时段的潮流。在这些场景中，关注发电的随机场景的发电量低于确定性场景，而关注电力系统福利的随机场景的发电量高于确定性场景。图 5(d - f) 展示了高峰时段的潮流。同样，在高峰时段，确定性场景的发电量介于关注电力系统福利的随机场景和关注发电的场景之间。

6.5. Power system welfare

6.5. 电力系统福利

We present the load profile for both deterministic and stochastic scenarios, followed by the power system welfare that the operator can generate under these scenarios. Fig. 6(a) illustrates the optimal charging schedule throughout the day. The red area highlights the power grid peak hours while the green area highlights the off-peak hours. The Deterministic case and ASAP case share the same area under curves because they are optimized without considering uncertainty in charging energy needs. However, the Stochastic (W)/Stochastic (G) case considers the uncertainty in energy consumption and thus has a different total charging demand. With a smart charging schedule (Deterministic/Stochastic), the charging profile becomes more flattened. This occurs because the generation cost is a quadratic function of power generation, meaning that a higher generation significantly increases the cost. Therefore, a good strategy is to shift high demand to time slots with lower demand. When comparing peak hours (times of high external load) with off-peak hours (times of low external load), we observe that the difference between the four methods becomes more pronounced during off-peak hours. For the stochastic cases, charging demand is uncertain, leading the operator to either charge more or less than the actual demand. In the power system welfare-focused case, the operator aims to maximize the benefit obtained by using the generated electricity minus the generation cost. In contrast, the generation-focused case aims to minimize generation costs only. The power system welfare-focused strategy is more aligned with economic benefits, while the generation-focused approach considers environmental factors. We find that under uncertainty, operators tend to charge more to increase power system welfare, even when generating less power could satisfy demand and reduce carbon emissions. Fig. 6(b) shows the power system welfare change of Deterministic, Stochastic (Welfare), and Stochastic (Generation) compared to without any control in AM (6:00-9:00), MD (9:00-15:00), PM (15:00-19:00), RD (19:00-6:00). The bar with '+' patterns indicates the change is positive and '/' indicates the change is negative. We find the change of Deterministic, Stochastic (Welfare) and Stochastic (Generation) is most significant in AM and RD when there are more charging sessions. While Deterministic, Stochastic (Welfare) provides positive changes, Stochastic (Generation) will reduce the power system welfare to improve environmental benefits. The simulation incorporating uncertainty reveals that areas with lower weather variability can be adequately managed through deterministic optimization, ensuring efficient charging operations. However, in regions with high weather-

related uncertainties-where travel time and energy consumption exhibit significant fluctuations-deterministic approaches may be insufficient. In such cases, aircraft must overcharge to compensate for unpredictable energy demands, leading to lower effective service rates and a potential reduction in overall network efficiency. This highlights the importance of choosing launch sites for UAM services in areas with stable operational conditions. Additionally, it highlights the need for designing a UAM network that is resilient to external uncertainties, such as weather disturbances and unexpected delays.

我们展示了确定性和随机场景下的负荷曲线，以及运营商在这些场景下可以产生的电力系统福利。图 6(a) 展示了全天的最优充电计划。红色区域突出显示电网高峰时段，而绿色区域突出显示非高峰时段。确定性情况和尽快充电 (ASAP) 情况在曲线下的面积相同，因为它们在优化时没有考虑充电能量需求的不确定性。然而，随机 (加权)/随机 (绿色) 情况考虑了能源消耗的不确定性，因此总充电需求不同。通过智能充电计划 (确定性/随机)，充电曲线变得更加平缓。这是因为发电成本是发电功率的二次函数，这意味着更高的发电量会显著增加成本。因此，一个好的策略是将高需求转移到需求较低的时段。当比较高峰时段 (外部负荷高的时段) 和非高峰时段 (外部负荷低的时段) 时，我们观察到四种方法之间的差异在非高峰时段变得更加明显。对于随机情况，充电需求是不确定的，这导致运营商要么充电多于实际需求，要么少于实际需求。在以电力系统福利为重点的情况下，运营商旨在最大化使用发电量减去发电成本所获得的收益。相比之下，以发电为重点的情况仅旨在最小化发电成本。以电力系统福利为重点的策略更符合经济效益，而以发电为重点的方法考虑环境因素。我们发现，在不确定性情况下，即使减少发电量可以满足需求并减少碳排放，运营商也倾向于多充电以提高电力系统福利。图 6(b) 显示了与上午 (6:00 - 9:00)、中午 (9:00 - 15:00)、下午 (15:00 - 19:00)、夜间 (19:00 - 6:00) 无任何控制时相比，确定性、随机 (福利) 和随机 (发电) 情况下的电力系统福利变化。带有 '+' 图案的条表示变化为正，'/' 表示变化为负。我们发现，在上午和夜间充电时段较多时，确定性、随机 (福利) 和随机 (发电) 的变化最为显著。虽然确定性、随机 (福利) 带来正向变化，但随机 (发电) 会降低电力系统福利以提高环境效益。纳入不确定性的模拟表明，天气变化较小的地区可以通过确定性优化进行充分管理，确保高效的充电运营。然而，在与天气相关的不确定性较高的地区——出行时间和能源消耗波动较大的地区——确定性方法可能不足。在这种情况下，飞机必须过度充电以补偿不可预测的能源需求，导致有效服务率降低，进而可能降低整体网络效率。这突出了在运行条件稳定的地区选择城市空中交通 (UAM) 服务发射站点的重要性。此外，这突出了设计一个能够抵御外部不确定性 (如天气干扰和意外延误) 的 UAM 网络的必要性。

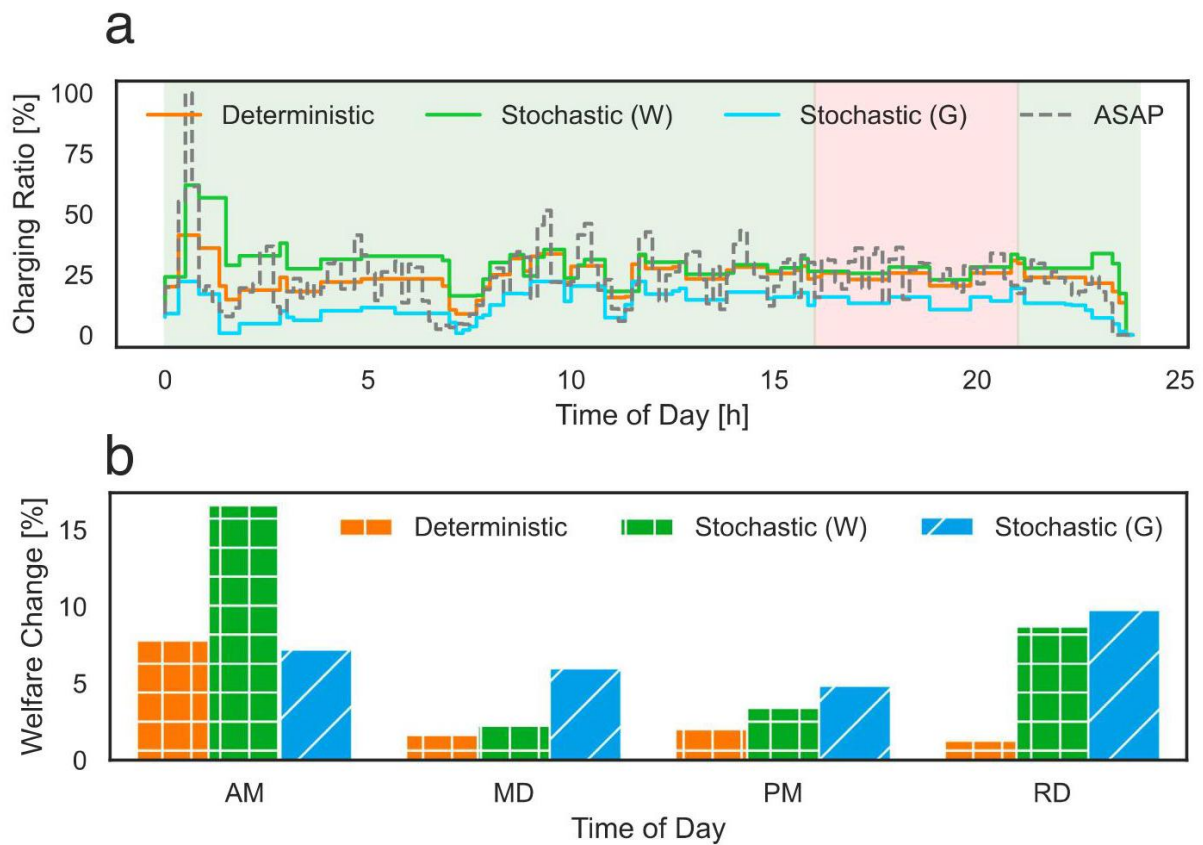


Fig. 6. Charging power and power system welfare. (a) Charging schedule under ASAP, Deterministic, Stochastic (power system welfare), and Stochastic (Generation). (b) power system welfare change of Deterministic, Stochastic (power system welfare), and Stochastic (Generation) compared with ASAP. Patch '+' denotes the positive change and '/' denotes negative changes.

图 6. 充电功率和电力系统福利。(a) 尽快充电 (ASAP)、确定性、随机 (电力系统福利) 和随机 (发电) 情况下的充电计划。(b) 与尽快充电 (ASAP) 相比, 确定性、随机 (电力系统福利) 和随机 (发电) 情况下的电力系统福利变化。补丁 '+' 表示正向变化, '/' 表示负向变化。

Table A. 3

表 A. 3

Example of UAM trips selection.

城市空中交通 (UAM) 出行选择示例。

Trip ID	Travel distance [mile]	Travel time savings [min]	Trip rank	Trip selected
1	100	20	3	No
2	60	30	2	Yes
3	100	30	1	Yes

7. Conclusion

7. 结论

The electrical grid is one of the many infrastructure challenges that must be overcome to enable large-scale UAM operations. This paper provides a framework to estimate the potential of smart charging to improve power system welfare when integrating large-scale UAM into the power grid. We first estimate UAM demand from LBS data. Then we construct a transportation and power system nexus by solving a fleet dispatch and an optimal power flow problem. The overall aim is to maximize the power grid welfare by smart charging management. At last, we incorporate the uncertainty of the charging time window and charging demand into the formulation through chance-constraint optimization. The results indicate that smart charging effectively enhances power system welfare. Nevertheless, to incentivize vertiports to adhere to the operator's smart charging schedule, an appropriate pricing strategy must be established. Beyond smart charging, several alternatives for integrating Urban Air Mobility (UAM) into the power grid we could explore include: (1) implementing renewable energy options like solar and wind power to meet some of the UAM charging needs, (2) utilizing energy storage systems, such as battery storage located at vertiports, to mitigate demand fluctuations and ensure grid reliability, and (3) exploring vehicle-to-grid (V2G) capabilities to improve grid flexibility. UAM vehicles equipped with bidirectional charging systems could help stabilize the grid by discharging stored energy back into it during peak demand periods.

电网是大规模城市空中交通 (UAM) 运营必须克服的众多基础设施挑战之一。本文提供了一个框架, 用于估计在将大规模 UAM 集成到电网中时, 智能充电对改善电力系统福利的潜力。我们首先根据位置服务 (LBS) 数据估计 UAM 需求。然后, 通过解决机队调度和最优潮流问题, 构建交通与电力系统的联系。总体目标是通过智能充电管理最大化电网福利。最后, 我们通过机会约束优化将充电时间窗口和充电需求的不确定性纳入模型。结果表明, 智能充电有效地提高了电力系统福利。然而, 为了激励垂直机场遵守运营商的智能充电计划, 必须制定适当的定价策略。除了智能充电, 我们可以探索的将城市空中交通 (UAM) 集成到电网中的其他几种方法包括:(1) 实施太阳能和风能等可再生能源选项, 以满足部分 UAM 充电需求;(2) 利用储能系统, 如位于垂直机场的电池储能, 以减轻需求波动并确保电网可靠性;(3) 探索车辆到电网 (V2G) 能力, 以提高电网灵活性。配备双向充电系统的 UAM 车辆可以在需求高峰期将储存的能量回馈电网, 从而帮助稳定电网。

CRedit authorship contribution statement

CRedit 作者贡献声明

Jiaman Wu: Writing - review & editing, Writing - original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Shangqing Cao: Writing - review & editing, Methodology, Data curation, Conceptualization. Mark Hansen: Supervision, Methodology, Investigation, Conceptualization. Marta C. González: Writing - review & editing, Supervision, Methodology, Investigation, Conceptualization.

吴佳曼: 写作 - 审阅与编辑、撰写初稿、可视化、方法学、调查、形式分析、数据管理、概念化。曹尚清: 写作 - 审阅与编辑、方法学、数据管理、概念化。马克·汉森: 监督、方法学、调查、概念化。玛尔塔·C·冈萨雷斯: 写作 - 审阅与编辑、监督、方法学、调查、概念化。

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我们感谢位置情报与测量平台 Cuebiq 提供汇总的出行数据。这些第一方数据是从选择通过符合 CCPA 和 GDPR 的框架匿名提供其位置数据访问权限的匿名用户那里收集的。通过 Cuebiq 的社会影响计划，Cuebiq 为学术研究和人道主义倡议提供出行洞察。Cuebiq 的负责任数据共享框架通过提供对可审计的本地沙盒环境的访问权限，使社会影响合作伙伴能够查询匿名和隐私增强的数据。提供给合作伙伴的所有最终输出都是汇总的，以保护隐私。

Appendix. Example of UAM demand sampling

附录。城市空中交通需求抽样示例

As shown in Table A.3, we have three trips; each trip is associated with a travel distance and travel time savings. Here, we set travel time savings as the most important factor and travel distance as the secondary important factor. After ranking, we select the first two trips as UAM trips. Therefore, we have Trip 2 and Trip 3 as UAM trips. This method can (1) select UAM trips from all trips given N^{trips} so as to include, and (2) be extended to include more factors when more data is available.

如表 A.3 所示，我们有三次出行；每次出行都与出行距离和节省的出行时间相关。在这里，我们将节省的出行时间设定为最重要的因素，将出行距离设定为次要因素。经过排序后，我们选择前两次出行作为城市空中交通出行。因此，我们将出行 2 和出行 3 作为城市空中交通出行。这种方法可以 (1) 从给定 N^{trips} 的所有出行中选择城市空中交通出行，以便纳入，并且 (2) 在有更多数据时扩展以纳入更多因素。

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